



EE4C10 : Fundamentals of Analog IC Design

Track: Bioelectronics Instructor: Dr. Dante Muratore

Topic: Sensing arrays

Q1/2024



Theme: Bioelectronics

- **Introduction to Bioelectronics at TU Delft**
- **Sensing arrays: applications and modalities**
 - Voltage
 - Current
 - Charge
- **All about pixels (almost)**
 - Pixel readout architectures: passive, active, digital
 - Active pixels (voltage, current, charge, multimodal)
- **Assignment**



Learning objectives

By the end of this class, you will be able to:

- Describe key concepts of **electrical sensing** (for bio)
- Illustrate different **sensing modalities**
- Recognise **pixel amplifier** configurations
- Explore **pixel design** tradeoffs
- **Design and simulate** a pixel readout circuit for biomedical applications
- **Develop a quantitative a feeling** for pixel density, number of transistors, area, complexity



We won't cover

- Electrode, sensor, photodiode model
 - Off-pixel circuitry
 - Digital pixels
-
- No reference textbook on pixel array chips (some on CMOS image sensors) References are mainly IEEE journal papers & online lecture notes



Teaching objective

By the end of this class, you will be:

- **in love with bioelectronics**





Bioelectronics @TUDelft



Mission of the section Bioelectronics

- Long: to invent and design biosensors, bio-MEMS, electrodes, micro-system integration, microfabrication, and biomedical circuits and systems for interaction with the electrophysiology of cells, tissues, organs and the human body, by means of electrical, magnetic, optical and ultrasound stimulation and recording or sensing, for in-vitro, ex-vivo and in-vivo medical and clinical applications
- Short: **to invent and design electronic systems to interact with biological tissue for investigational and clinical purposes**

Professors in Bioelectronics



Wouter Serdijn: Bioelectronics,
Biomedical circuit and systems,
Bioelectronic medicine



Paddy French:
Biomedical devices,
Sensor technology



Frans Widdershoven:
CMOS smart biosensor ICs
and systems



Leon Abelmann:
Magnetostatics, 3D-printing,
Self-assembly, MEMS



Vasiliki Giagka:
Flexible implants, Microfabrication,
Microsystem integration



Tiago Costa: IC design and
microsystem integration
for electroceuticals



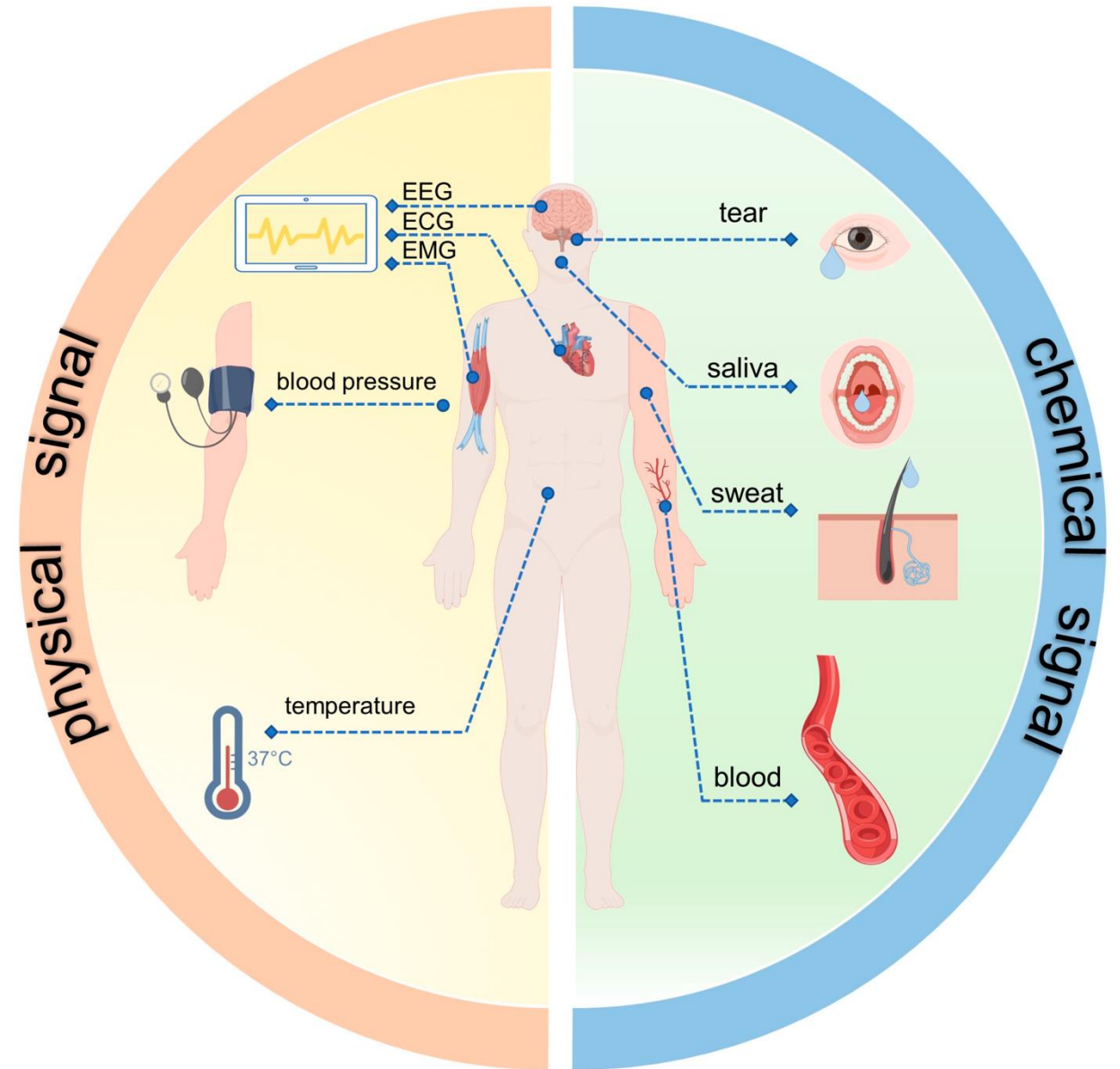
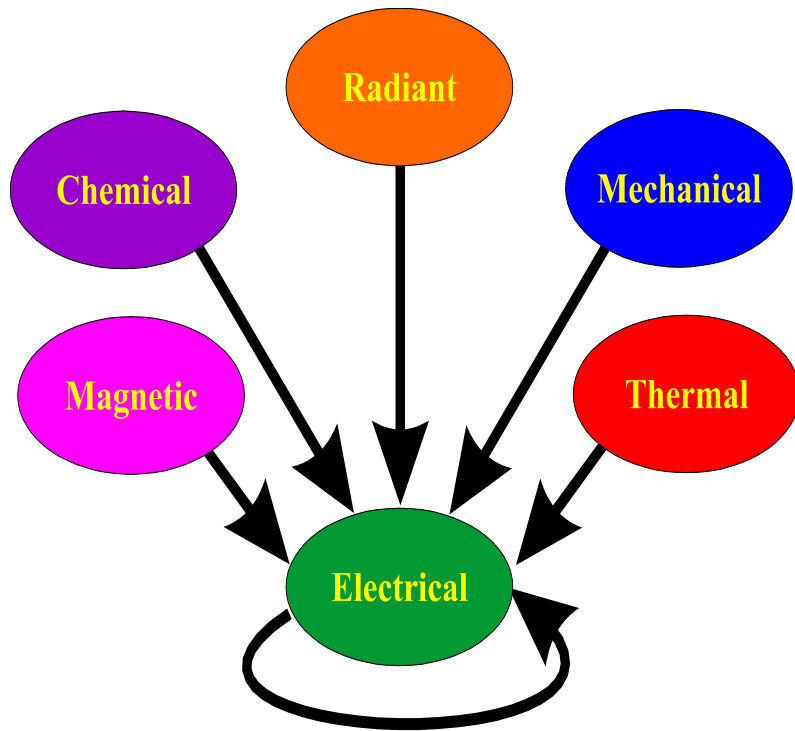
Dante Muratore: IC design for
brain-machine interfaces
and wearable sensors



Achilleas Savva:
Organic optoelectronics

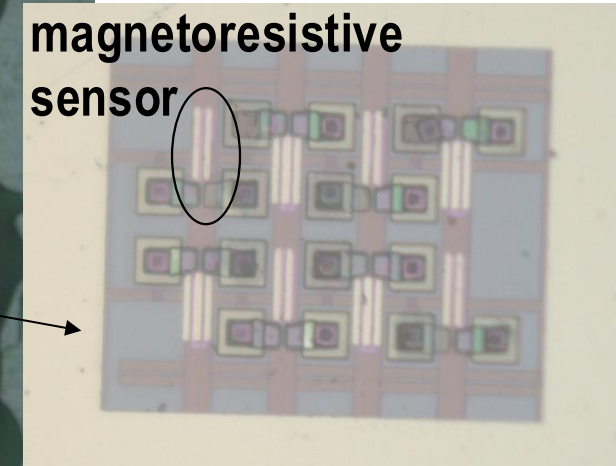
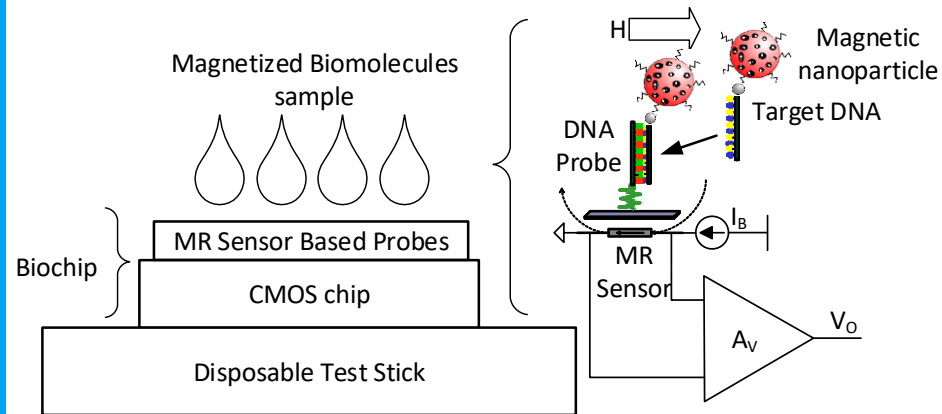


Biosensors



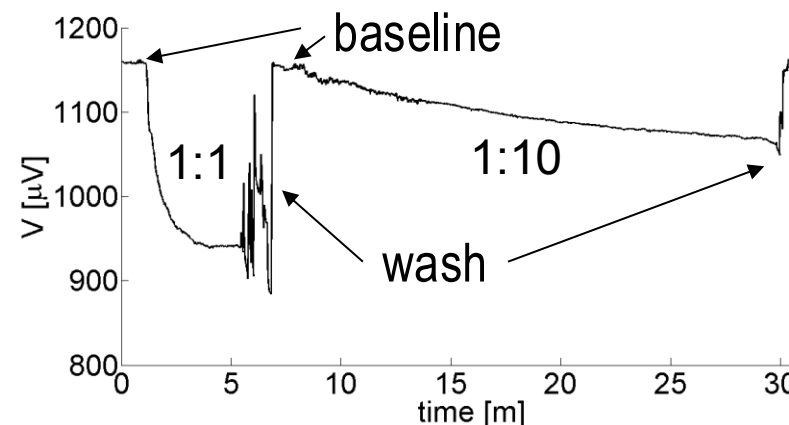


HIGH DENSITY MAGNETIC SENSOR ARRAYS ON CMOS FOR BIOMOLECULAR RECOGNITION DETECTION

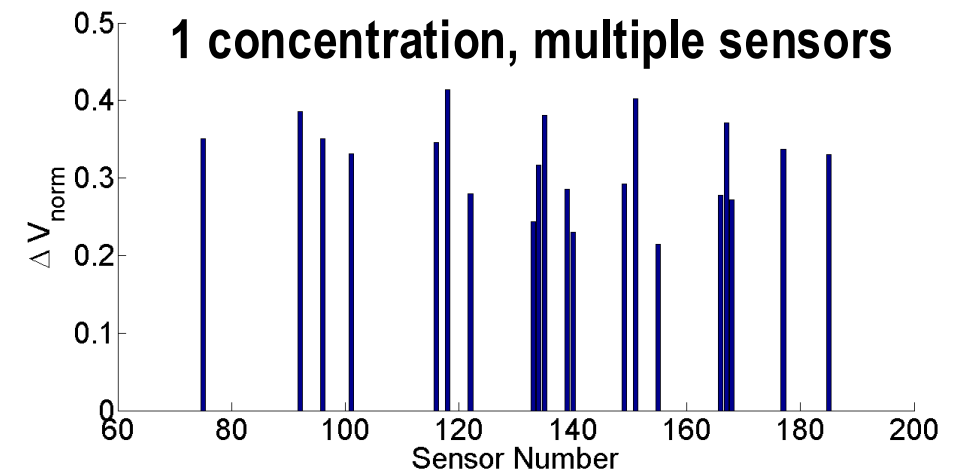


Measurement of magnetic bead concentration →

1 sensor, 2 concentrations



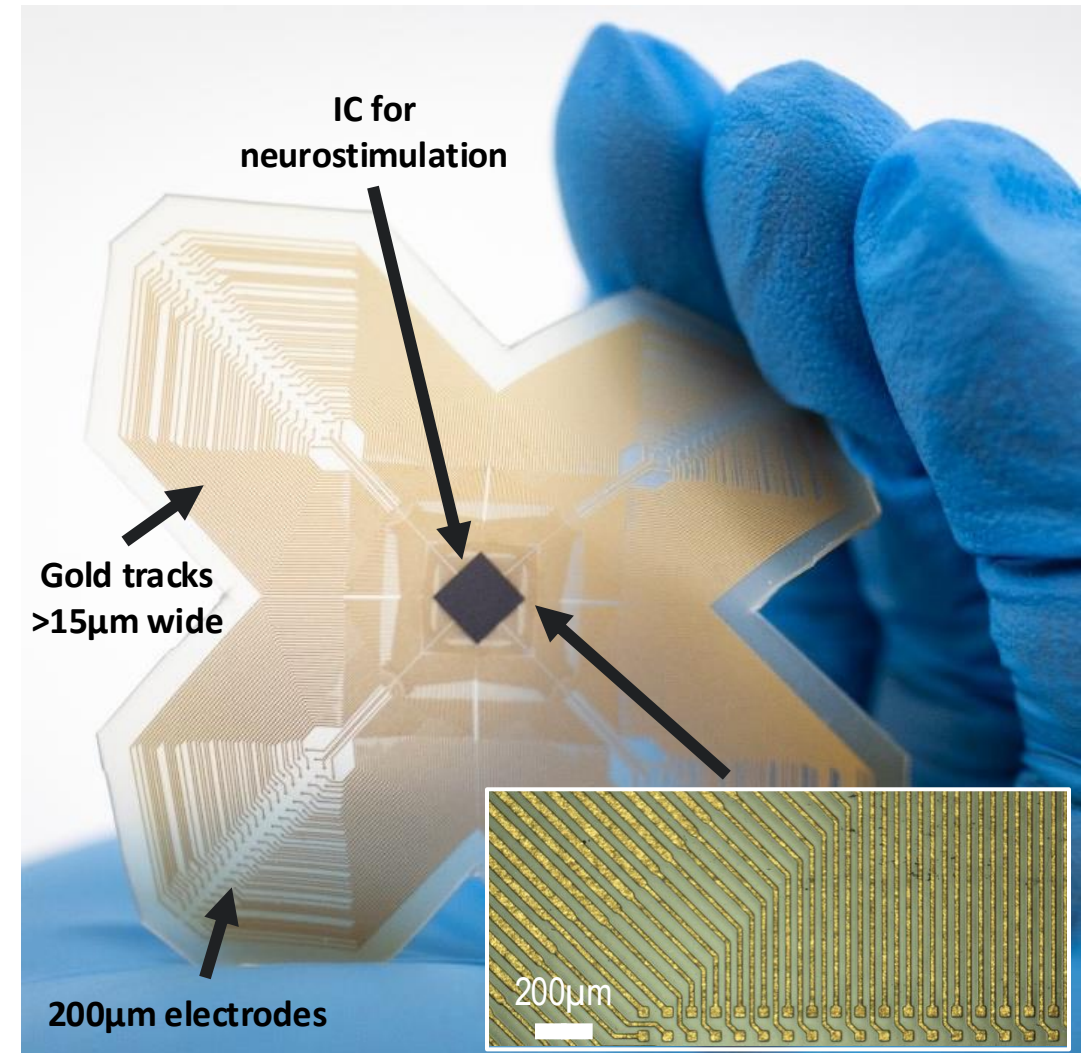
1 concentration, multiple sensors





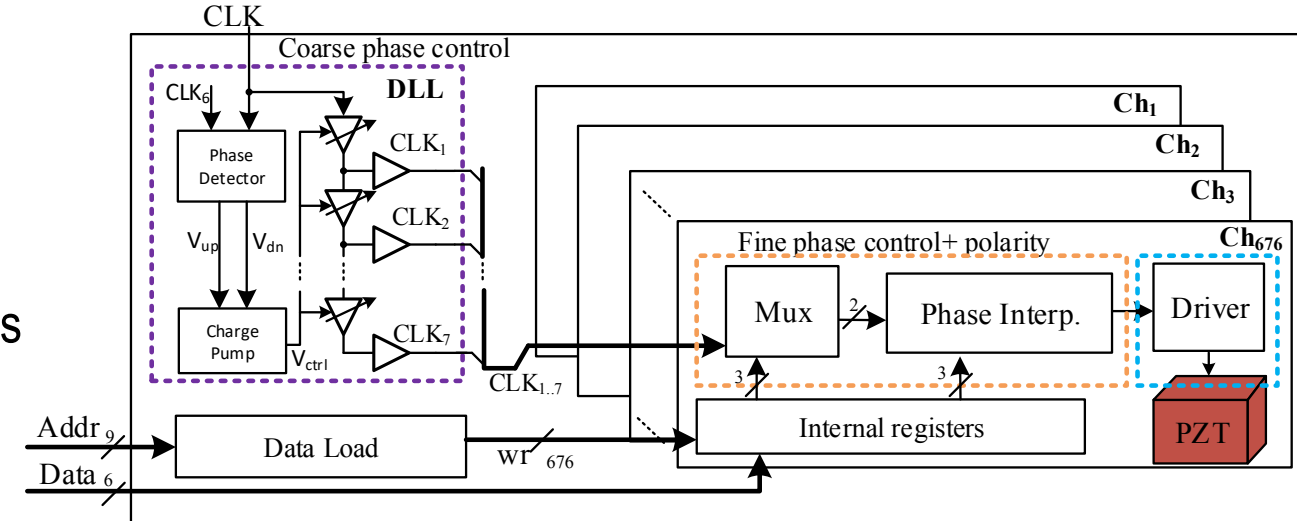
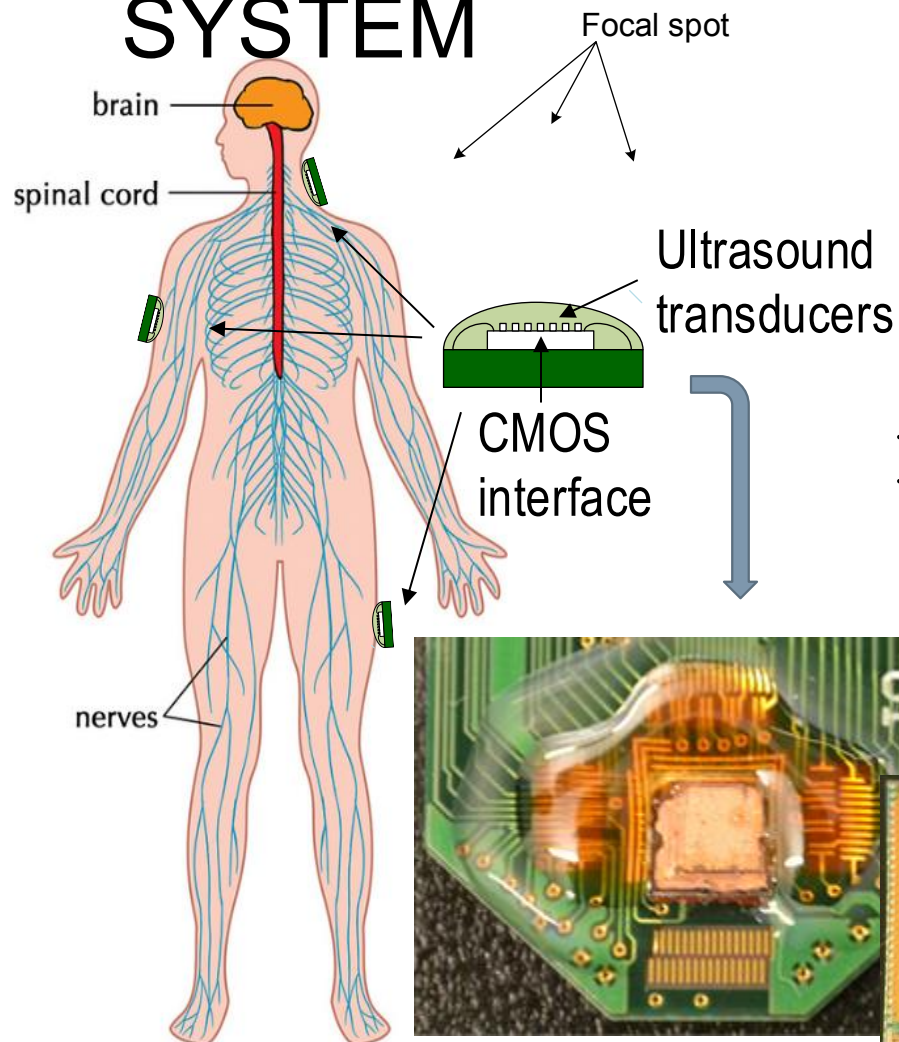
Towards flexible brain implants with thousands of independent channels

- Technology for high-density **neurostimulation and recording of the Central Nervous System**
- IC with possibility for **1008** neurostimulation **channels**
- **Fully-flexible** TPU implant with more than **324** recording or stimulating **electrodes** 200 μ m in diameter
- Optimized high-density interconnects down to 15 μ m in width

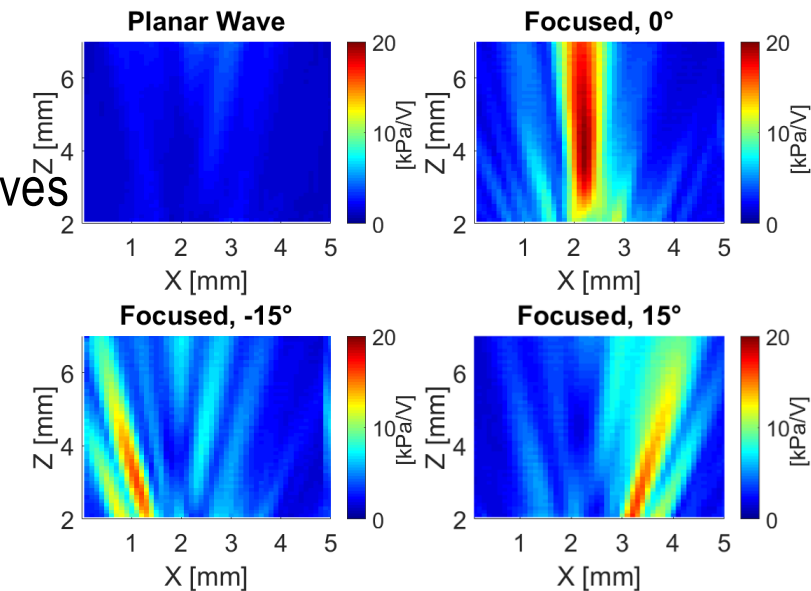




NON INVASIVE ULTRASOUND-BASED NEUROMODULATION OF THE NERVOUS SYSTEM

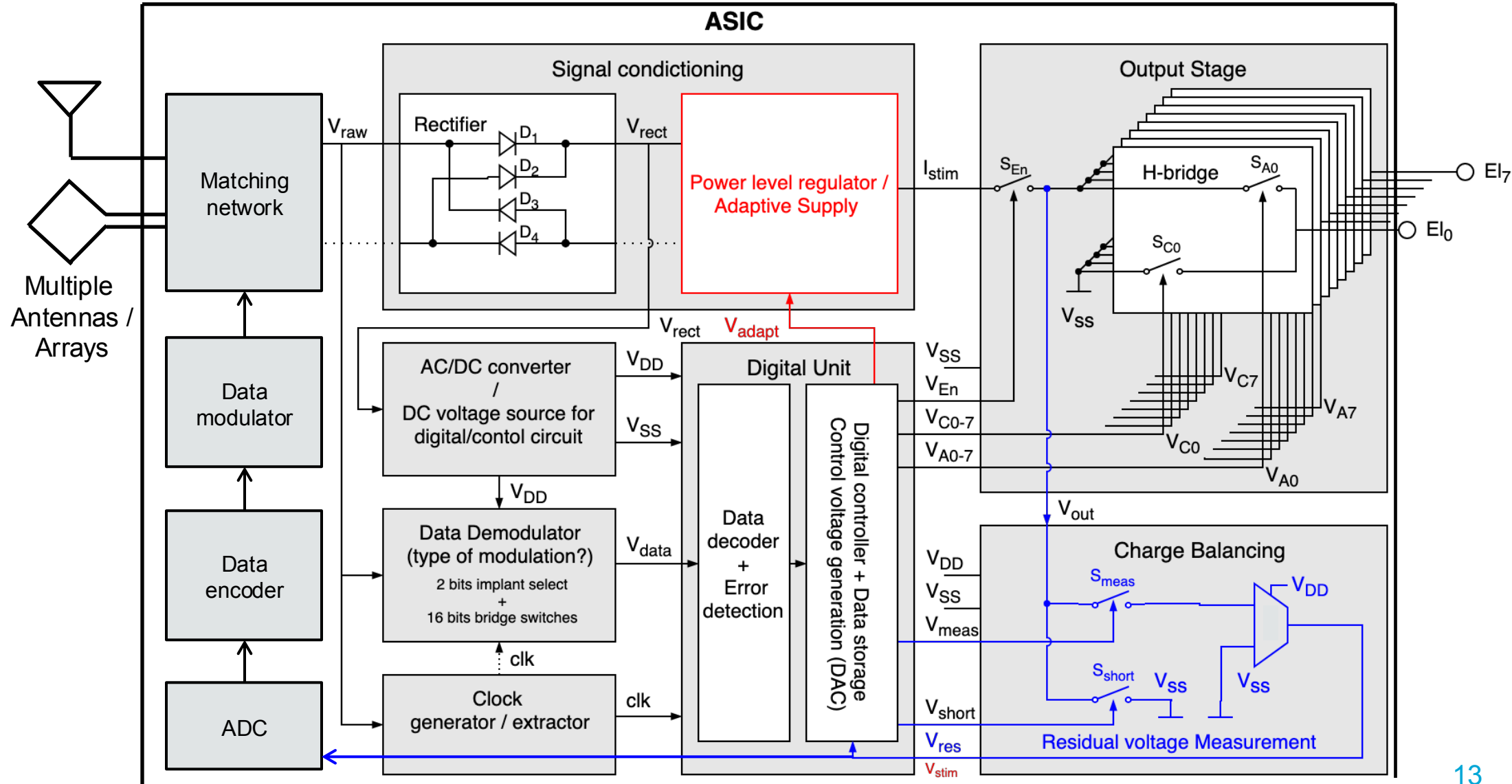


Steering and focusing ultrasound waves



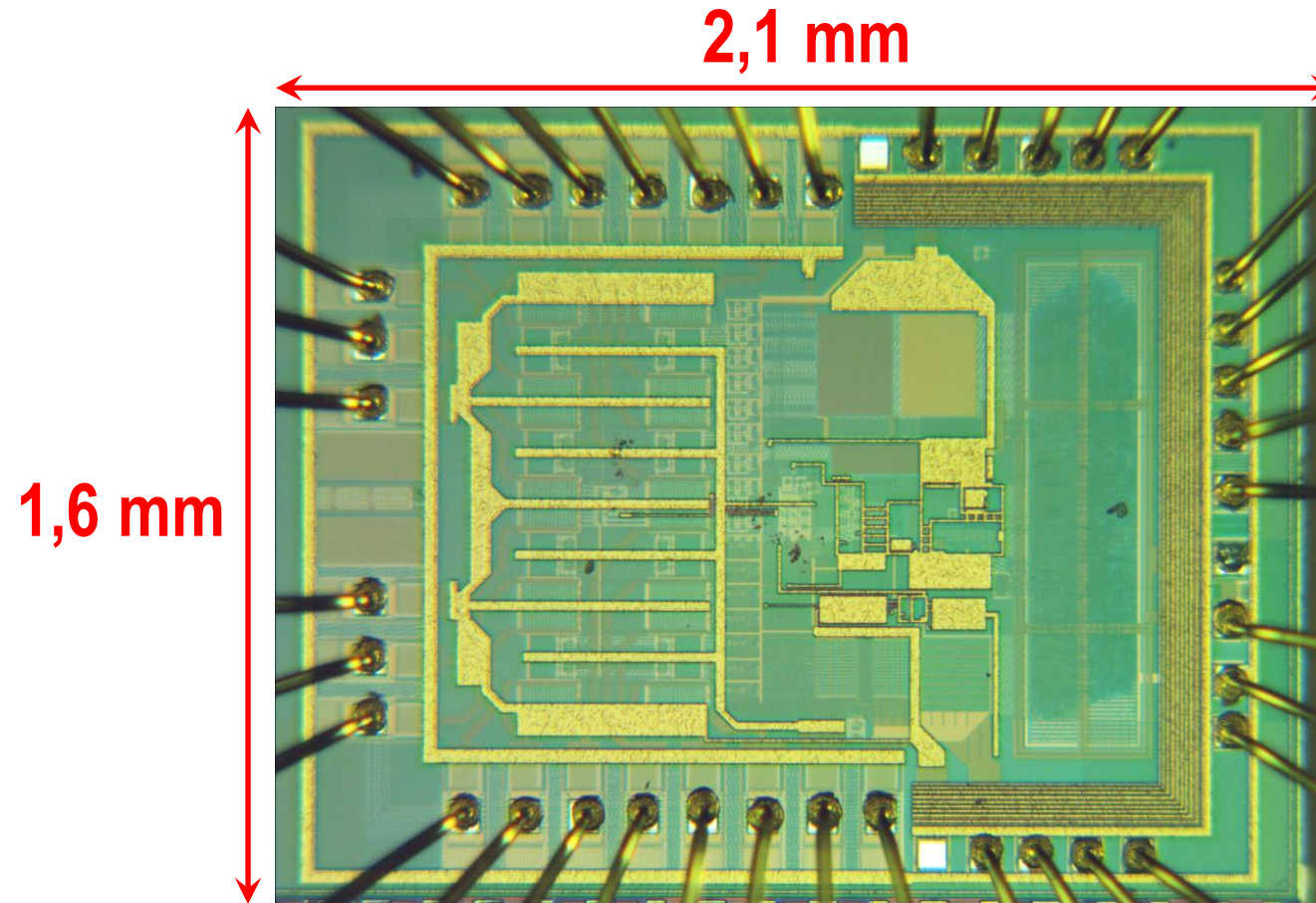


Complete Neural stimulator system level design





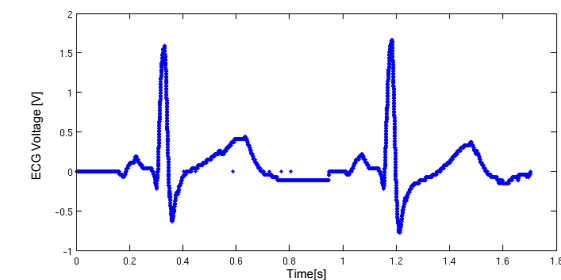
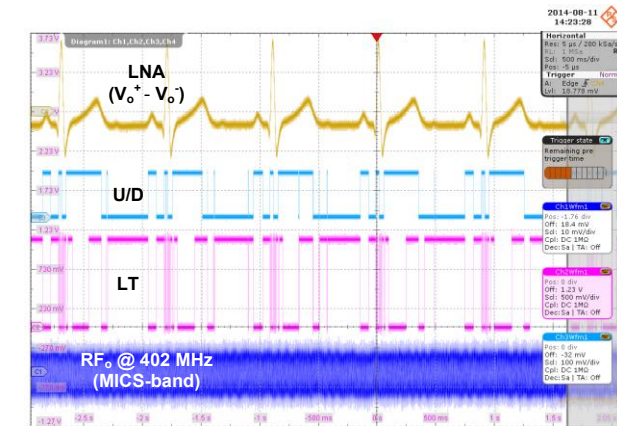
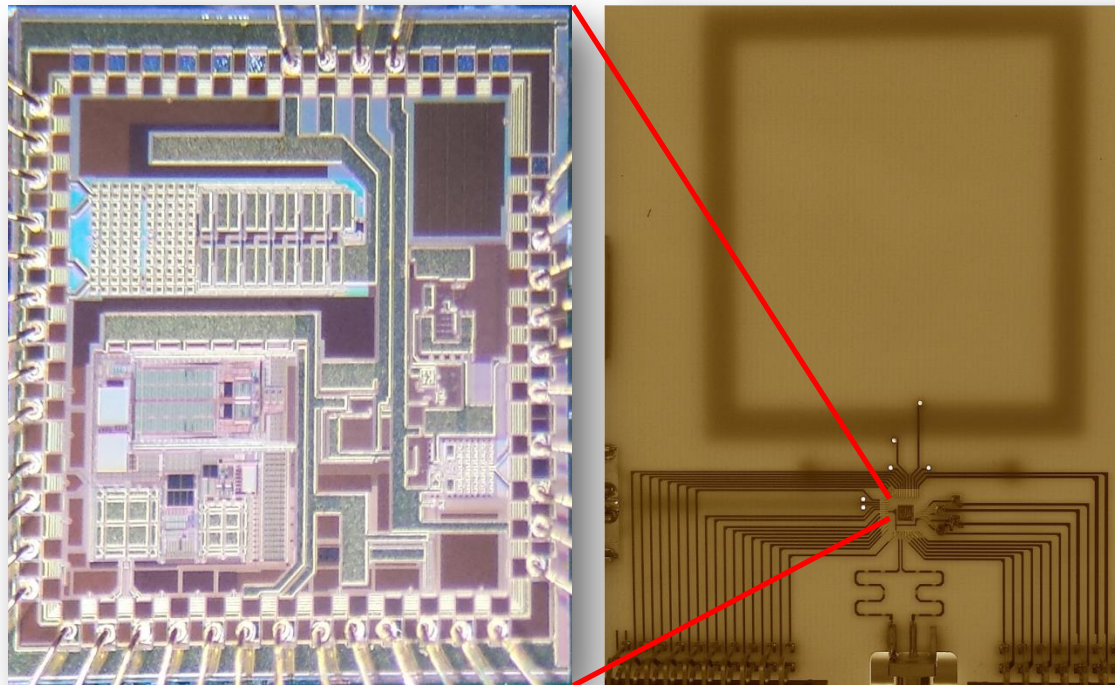
A 16-CHANNEL UHF NEUROSTIMULATOR IC WITH RECORD POWER EFFICIENCY



16-electrode 8-channel neurostimulator IC with unprecedented multi-channel stimulation power efficiency. Designed by Alessandro Urso and Marijn van Dongen



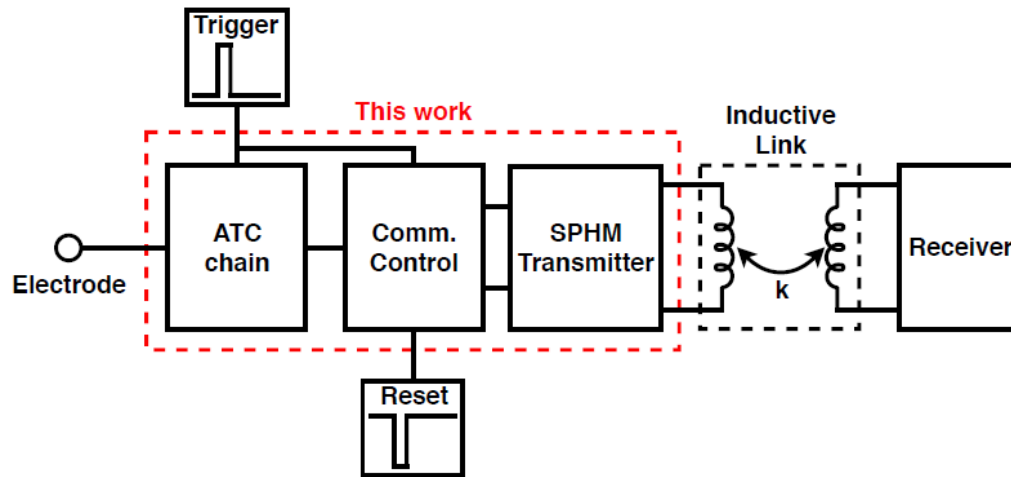
An Autonomous Wireless Sensor Node for ECG Monitoring



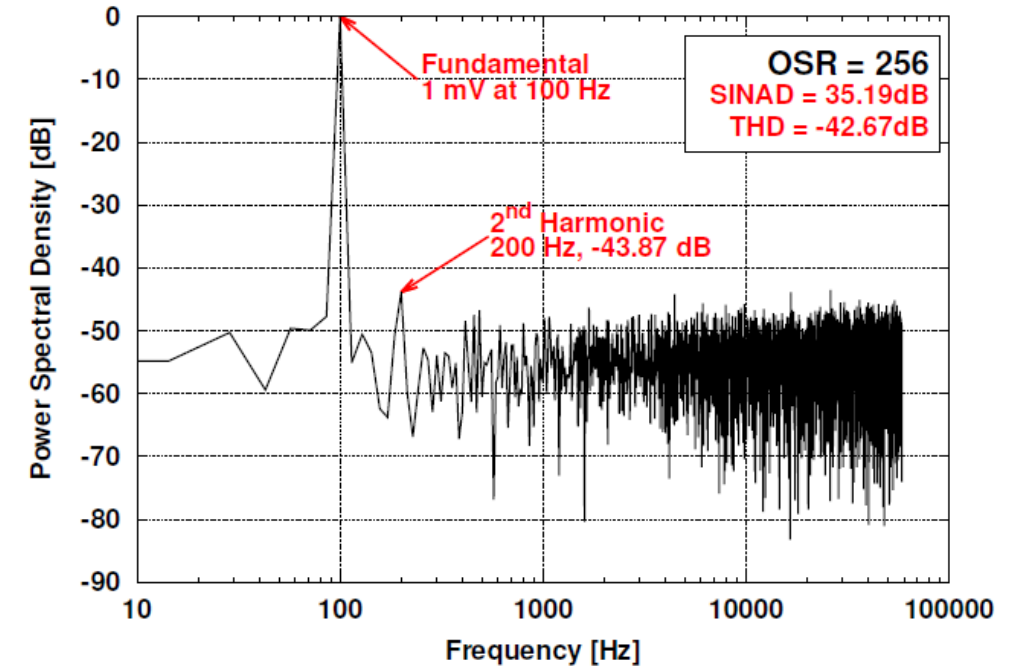
This tiny chip (left) when mounted on a printed circuit board (middle) with electrodes and antennas becomes a batteryless ECG tag allowing the readout and wireless transmission of cardiac signals. Designed by Andre Mansano and Yongjia Li. Support by CNPq and CSC



11 nW Signal Acquisition Platform for Remote Biosensing



Technology	$0.18\mu m$
V_{DD}	0.6 V
BW	1.48 kHz
Noise	$12.3\mu V_{rms}$
SINAD	35.19 dB
$P_{@256Hz}$	10.94 nW



Energy per block:

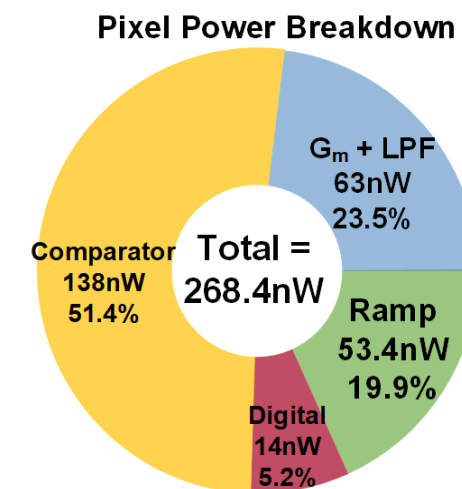
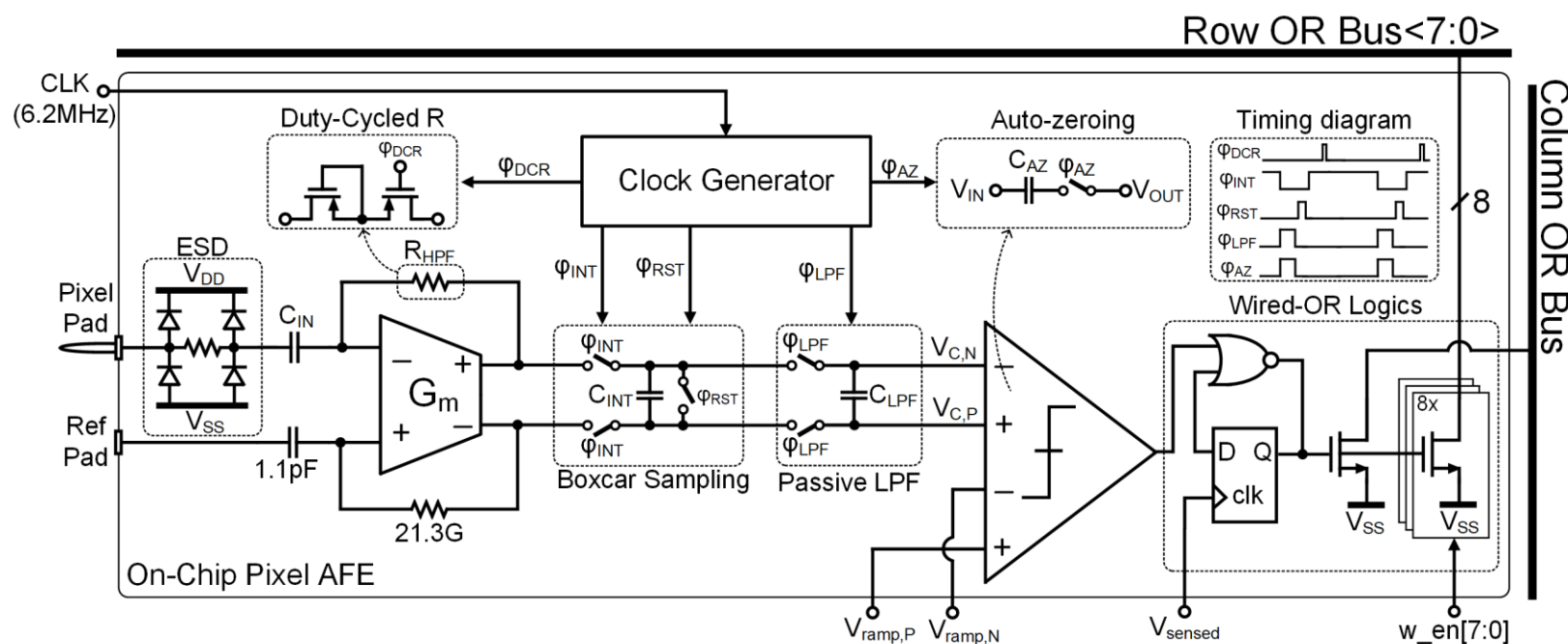
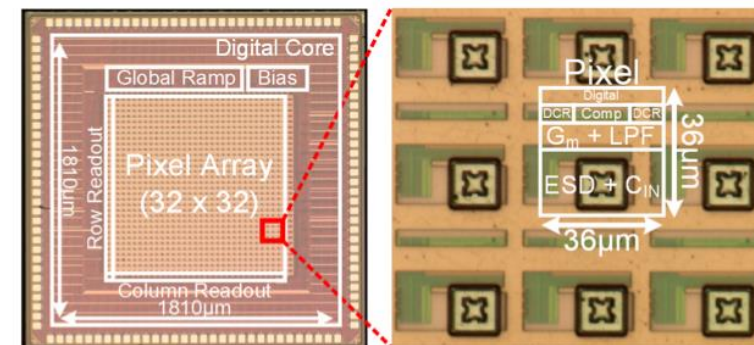
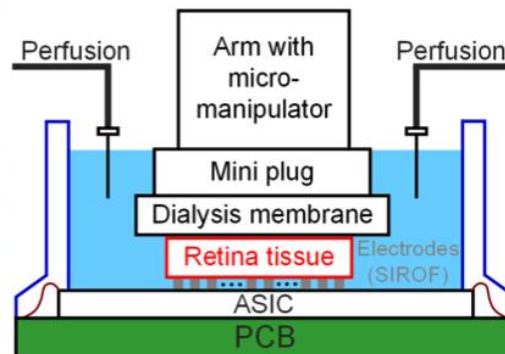
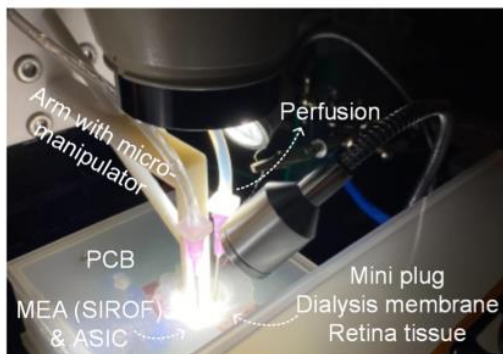
ATC Chain	2.19 pJ/sample
Interface	2.31 pJ/sample
TX	32.58 pJ/sample

Total energy dissipation:

42.72 pJ/sample

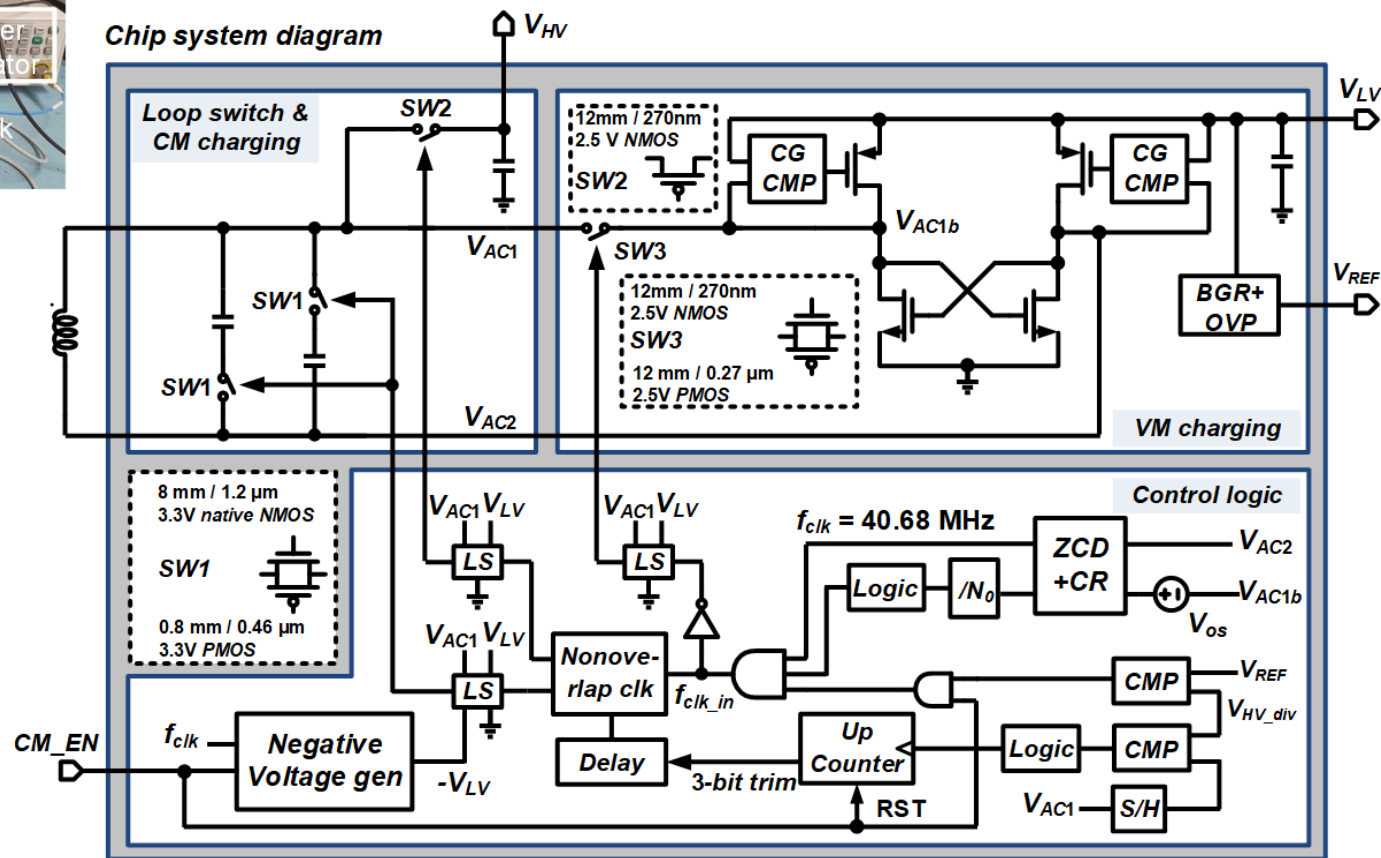
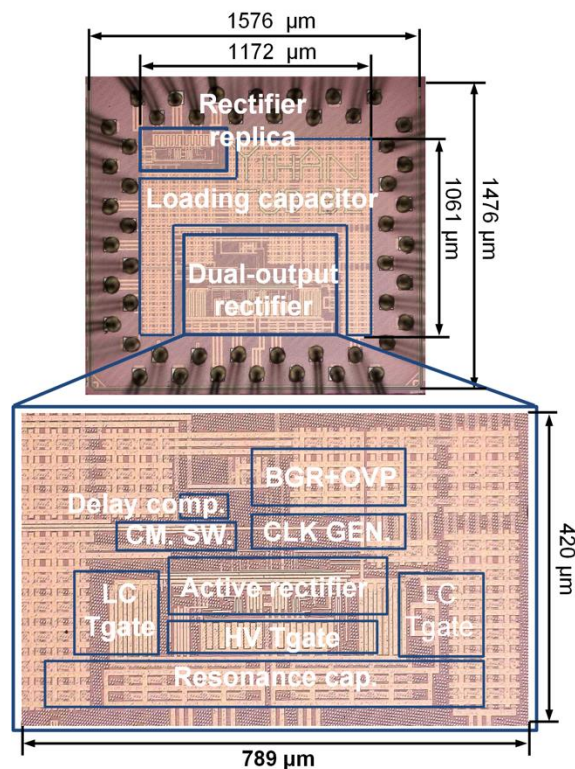
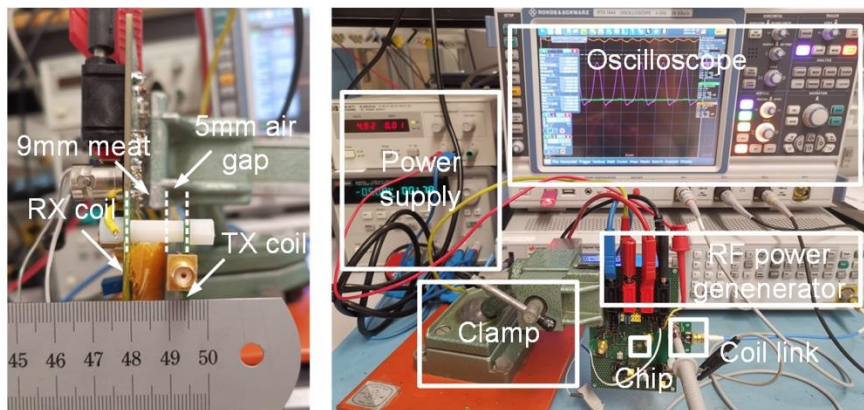


A 1024-Channel 268 nW/pixel 36x36 $\mu\text{m}^2/\text{ch}$ wired-OR ASIC





Inductive Wireless Power Transfer





RECOMMENDED COURSES FOCUS AREA

BIOMEDICAL CIRCUITS AND SYSTEMS

Q1 or Q5	Q2	Q3	Q4
BM41005 Anatomy and Physiology (4)		ET4127 Themes in Biomedical Electronics (4)	EE4555 Active Implantable Biomedical Microsystems (5)
EE4C08 Measurement and Instrumentation (5)	EE4257 Sensors and Actuators (4)	ET4252 Analog Integrated Circuit Design (4)	BM41060 Physiology and Engineering (3)
BM41095 Medical Instruments A: Clinical Challenges and Eng. Solutions (3)	EE4109 Structured Electronic Design (5)	ET4130 Bioelectricity (3)	ET4278 Over-Sampled Data Converters (4)
EE4C10 Analog Circuit Design Fundamentals (5)	EE4520 Analog CMOS Design 1 (3)	ET4369 Nyquist-Rate Data Converters (4)	ET4351 VLSI Systems on Chip (4)
EE4C04 Control Theory or EE4C07 Advanced Computing Systems (5)	EE4610 Digital IC Design (3)	EE4615 Digital IC Design II (3)	
WM1402TU Ethics of Healthcare Technologies (5)			



RECOMMENDED COURSES FOCUS AREA

BIOSENSORS, BIOMEMS AND MICROSYSTEM

Q1 or Q5	Q2	Q3	Q4
BM41005 Anatomy and Physiology (4)		ET4127 Themes in Biomedical Electronics (4)	EE4555 Active Implantable Biomedical Microsystems (5)
EE4C08 Measurement and Instrumentation (5)	EE4257 Sensors and Actuators (4)	EE4289 Integrated Circuits and MEMS Technology (4)	ET4260 Microsystem Integration (4)
BM41095 Medical Instruments A: Clinical Challenges & Eng. Solutions (3)		ET4130 Bioelectricity (3)	ET4icp IC Technology Lab (2)
WM1402TU Ethics of Healthcare Technologies (5)		ET4391 Advanced Microelectronics Packaging (3)	BM41060 Physiology and Engineering (3)
		ET4277 Microelectronics Reliability (4)	



MSc Thesis projects

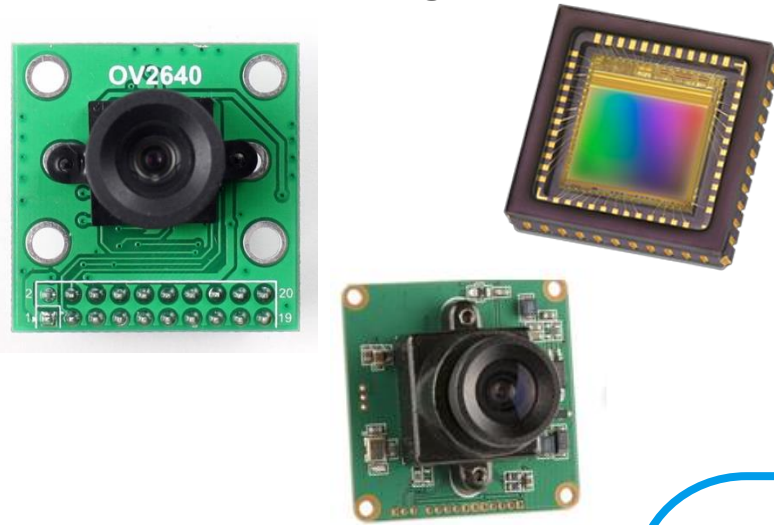
- Quite a few suggestions for projects you can find on our web site:
<http://bioelectronics.tudelft.nl/Education/mscprojects.php>
- But it is highly recommended to contact us and make an appointment to discuss a suitable thesis project



Sensing Array Applications

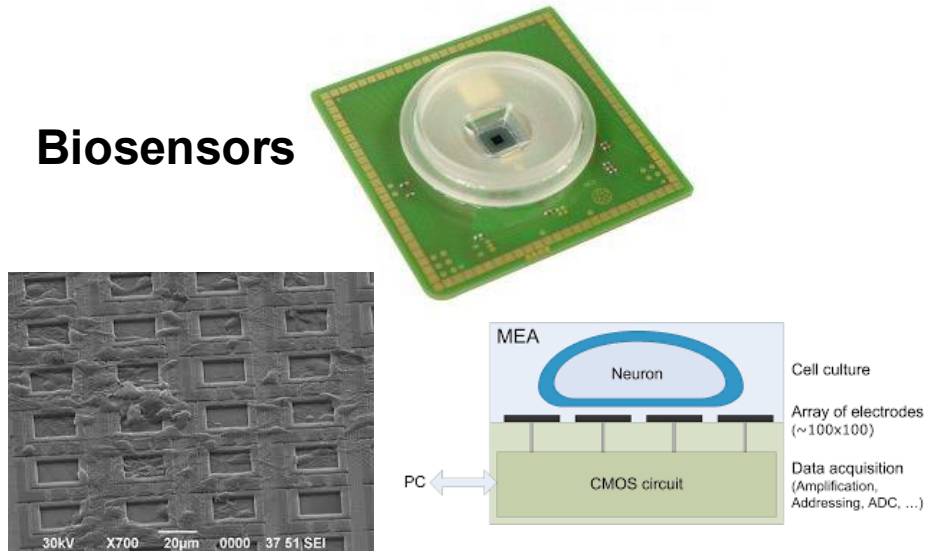


CMOS Imagers

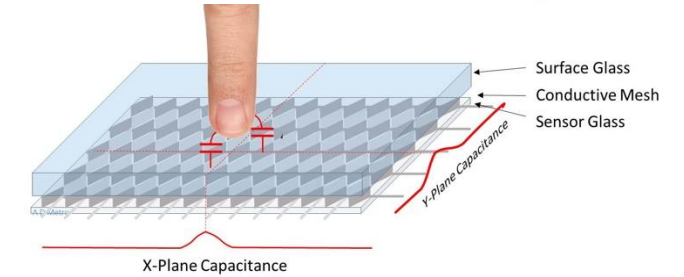
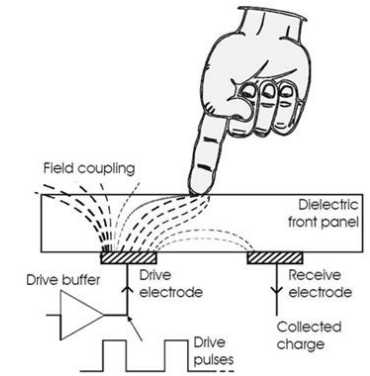


Pixel Array Chips

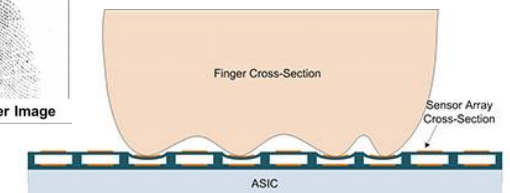
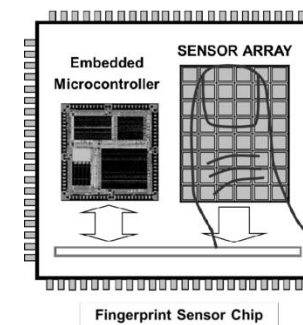
Biosensors



Touch screens

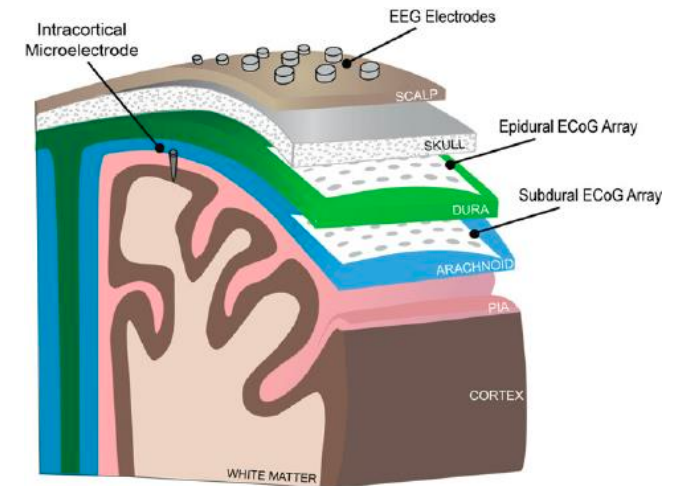
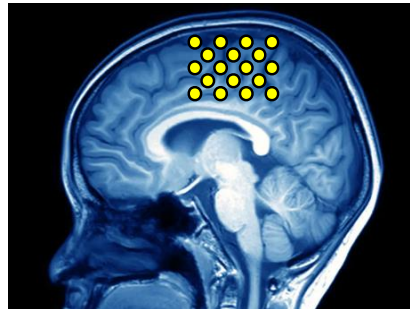


Fingerprint sensors



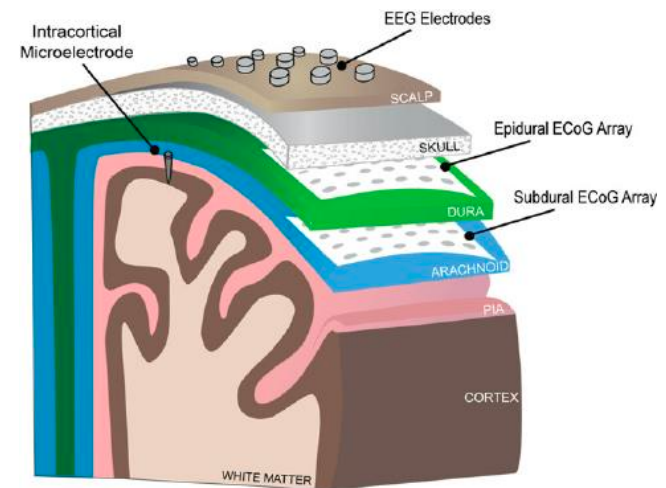
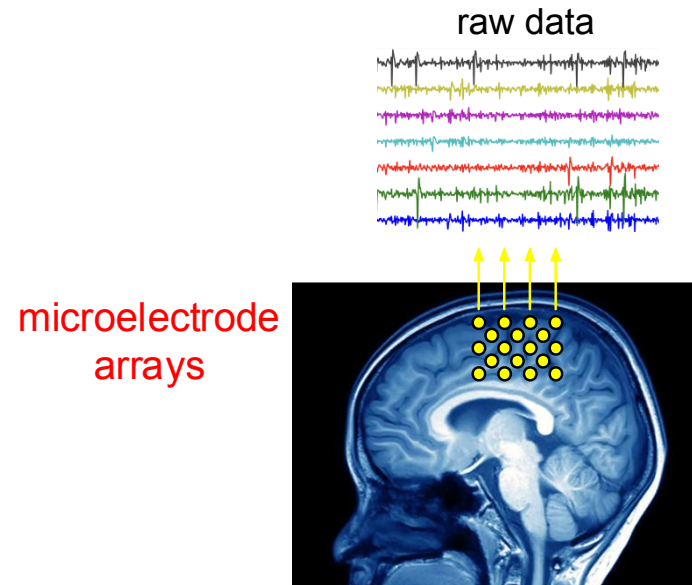


BMI: a link to the nervous system





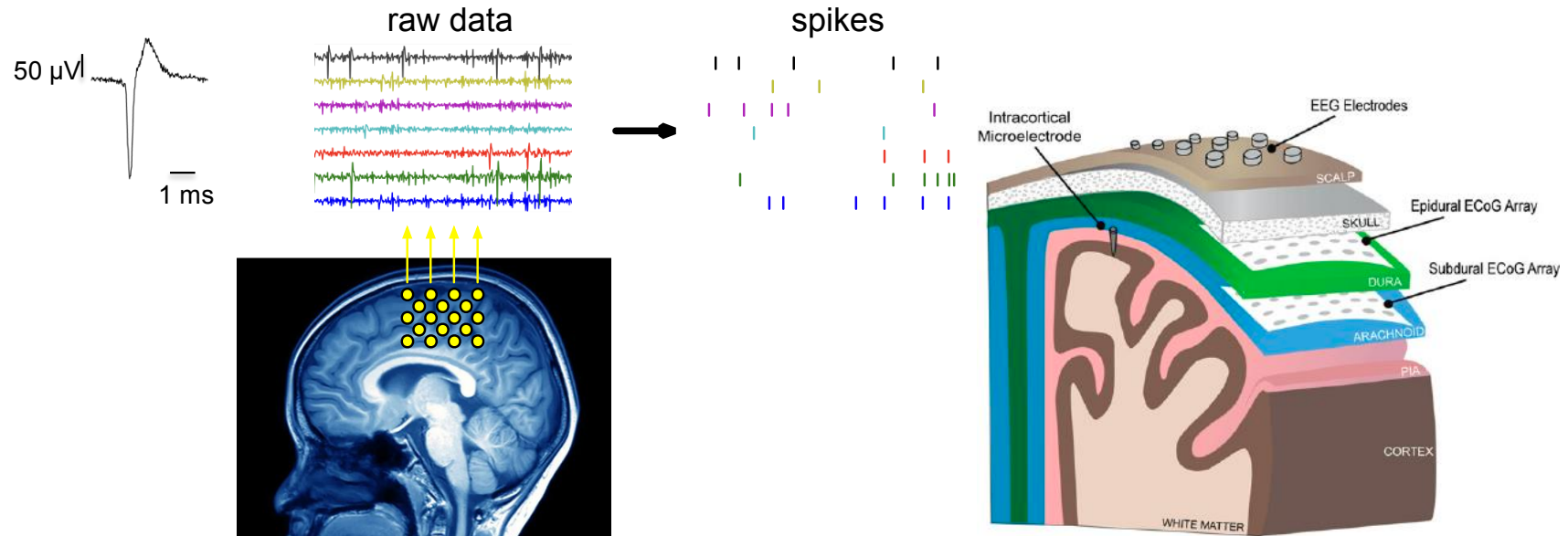
BMI: a link to the nervous system



- We are capable of externally recording (and stimulating) neurons in the electrical domain



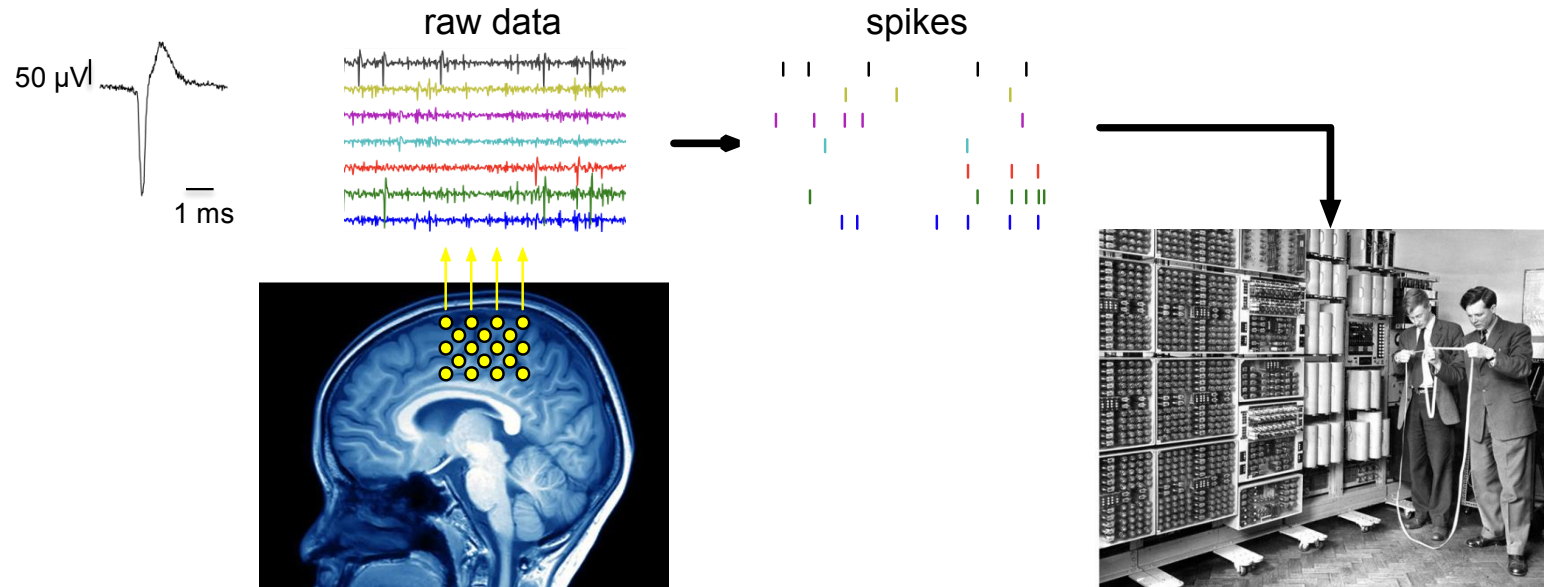
BMI: a link to the nervous system



- We are capable of externally recording (and controlling) neurons in the electrical domain
- Information in most neurons is encoded in spikes - perturbations in the voltage across the cell membrane



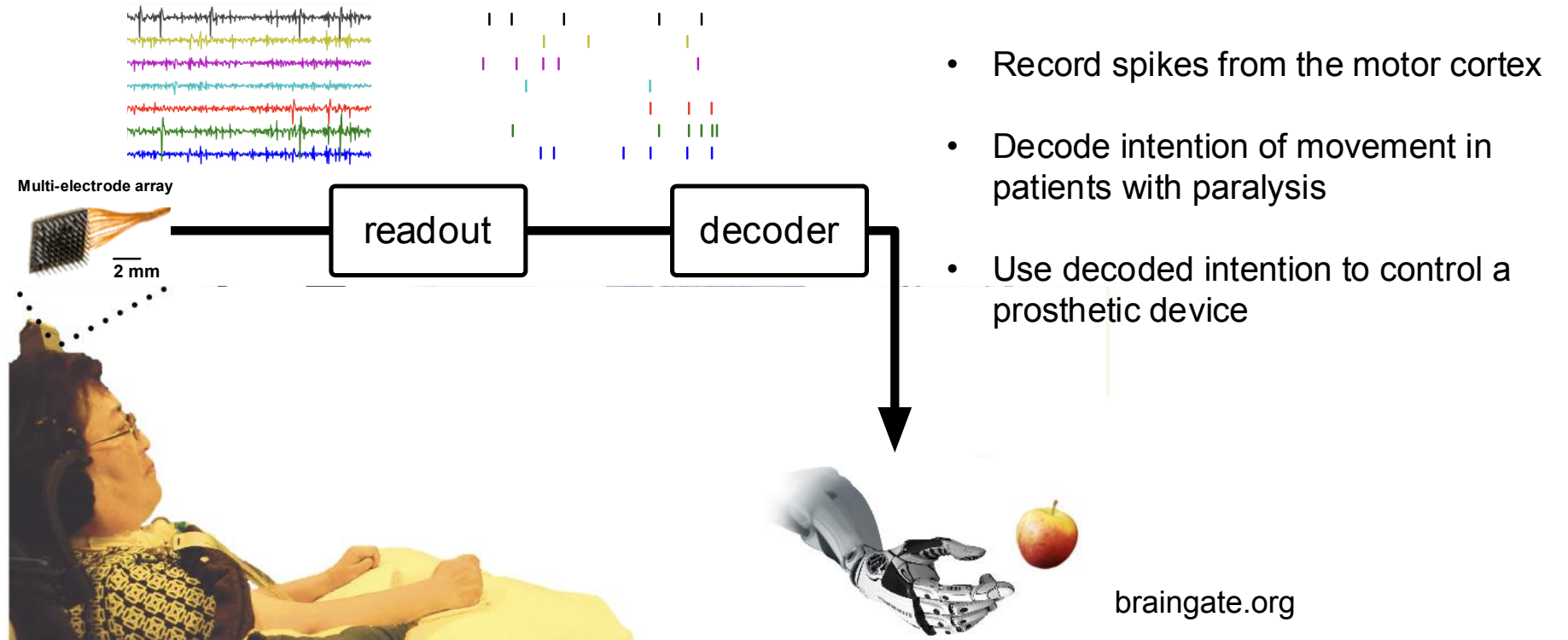
BMI: a link to the nervous system

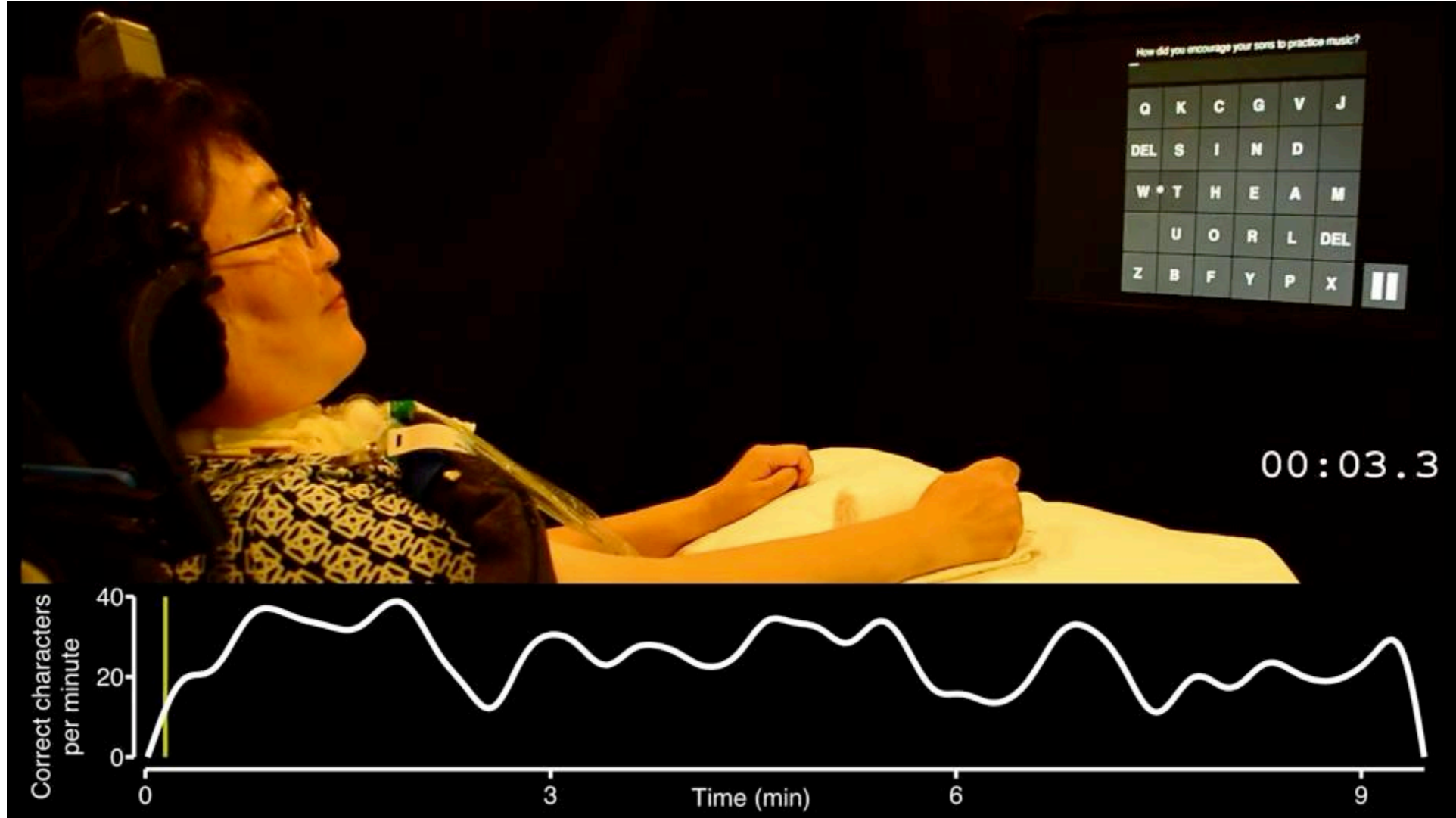


- We are capable of externally recording (and controlling) neurons in the electrical domain
- Information in most neurons is encoded in spikes - perturbations in the voltage across the cell membrane
- Make sense of this information to perform various tasks



Example: controlling a robotic arm



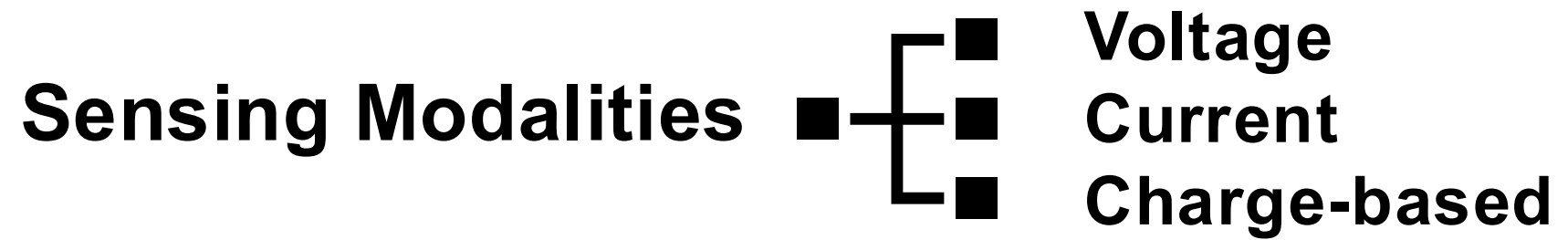




Sensing Modalities

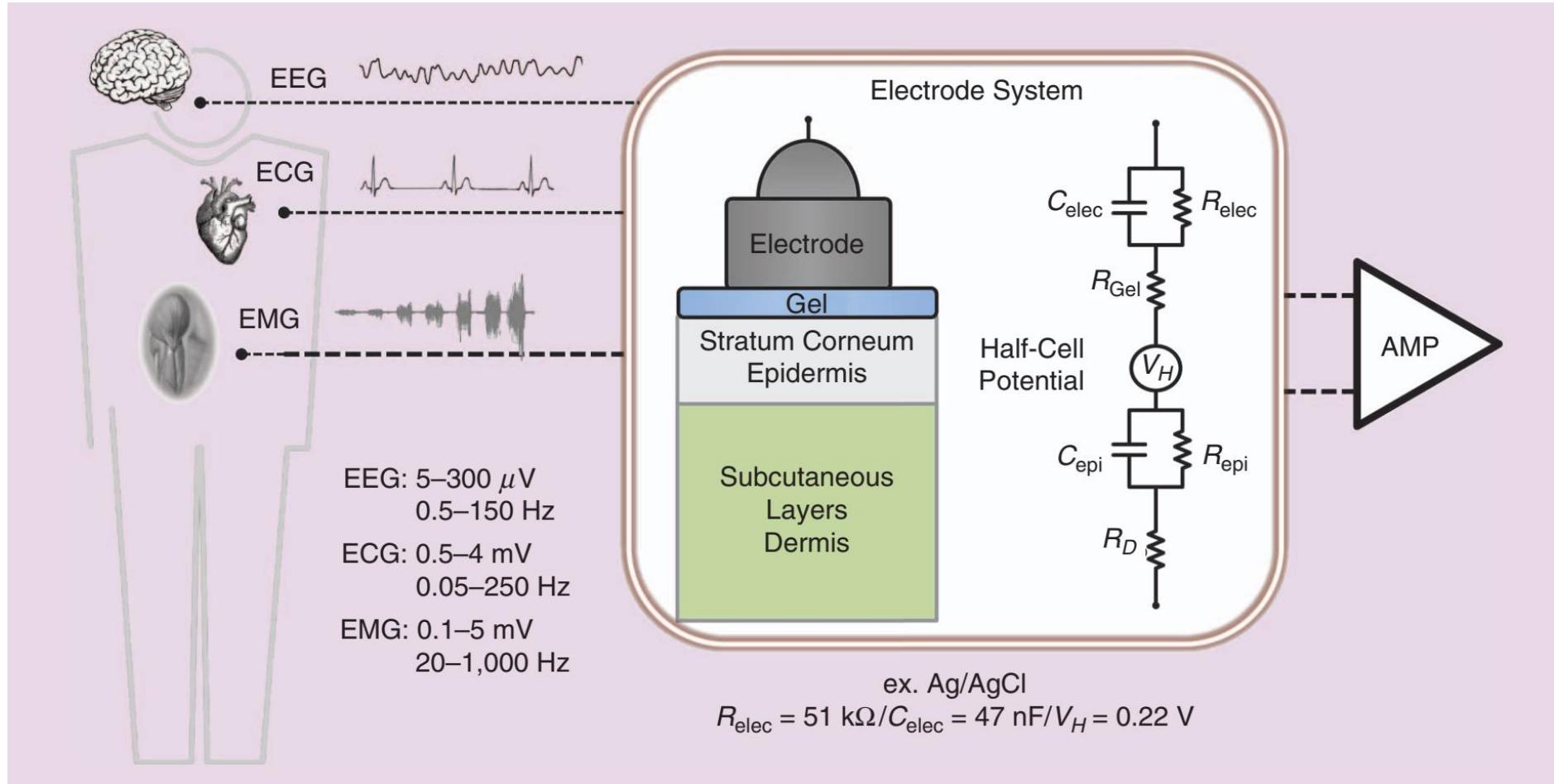


Background





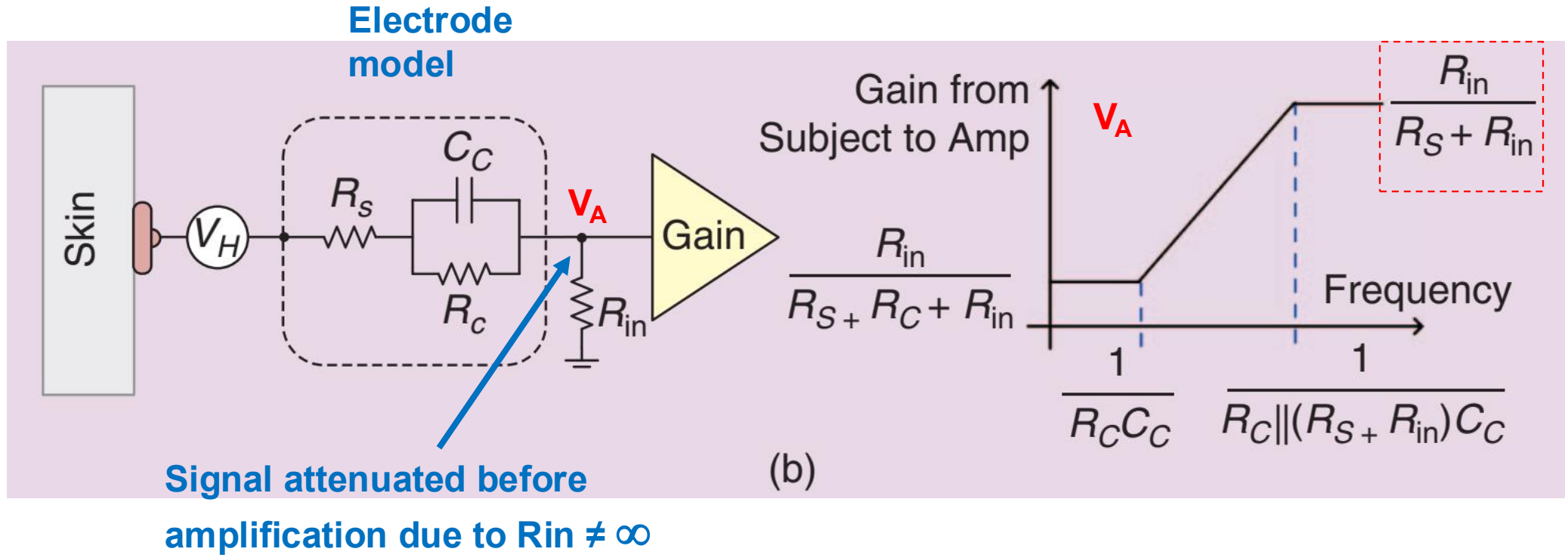
Voltage Sensing in Biosignal Acquisition Systems





Voltage Sensing

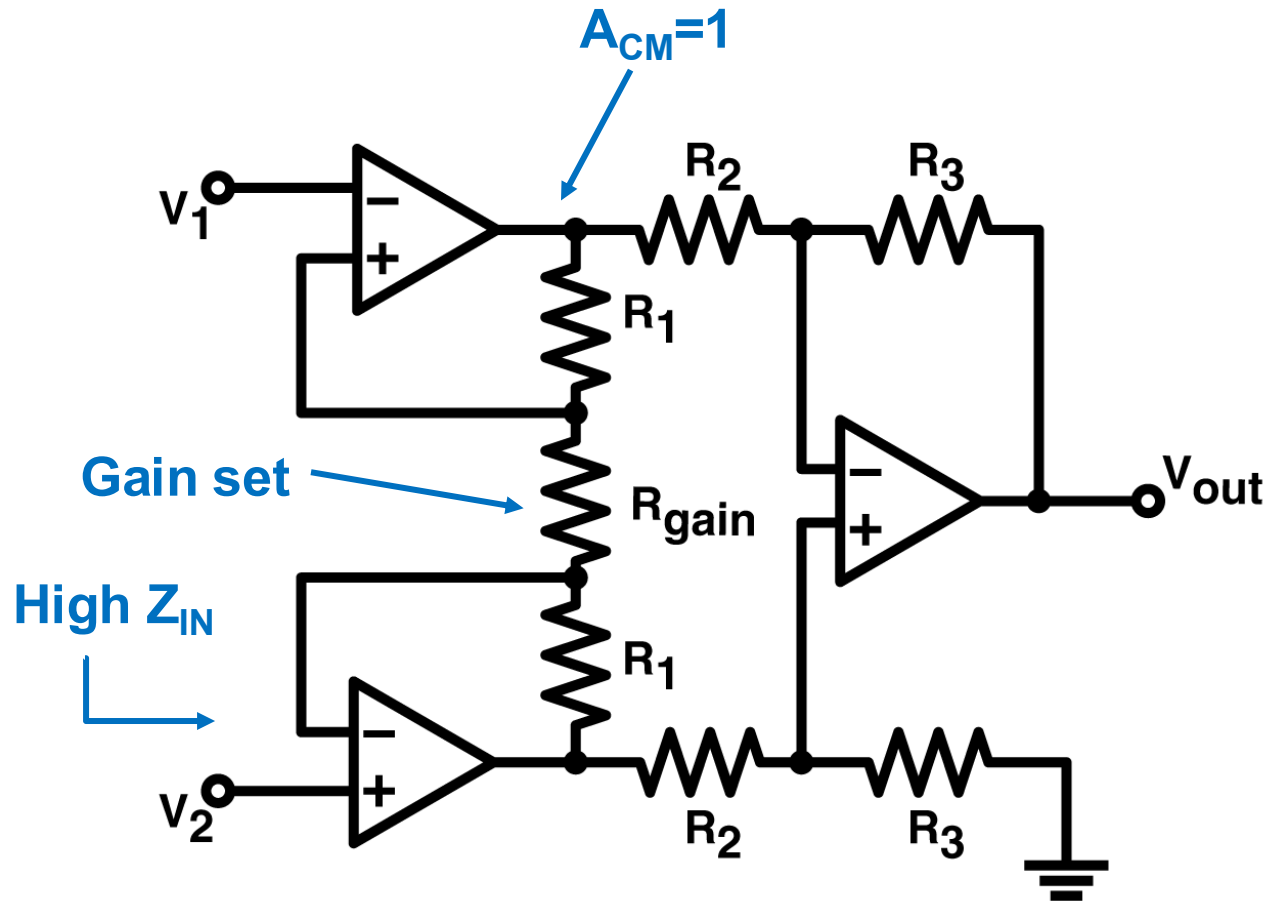
Electrode impedance





Voltage Sensing

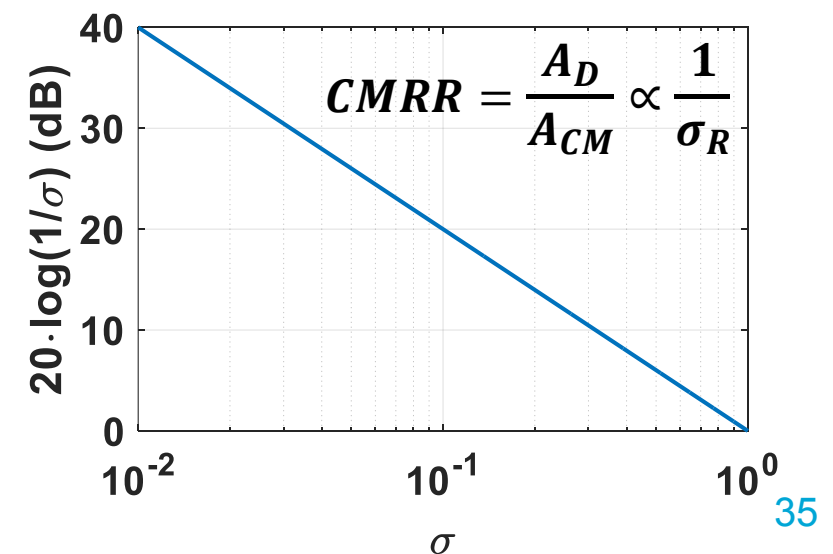
3-opamp



NOT SUITABLE FOR ASICs!

- CM transferred to the 2nd stage
- Accurate matching of resistor network → laser trimming
- CMRR ~ 80dB
- Output buffer to drive R's
- **Power! 3X I_{AMP}**

	First Stage	Second Stage
A_D	$1+(2R_1/R_{gain})$	R_3/R_2
A_{CM}	1	$\sigma_R \times R_3/R_2$

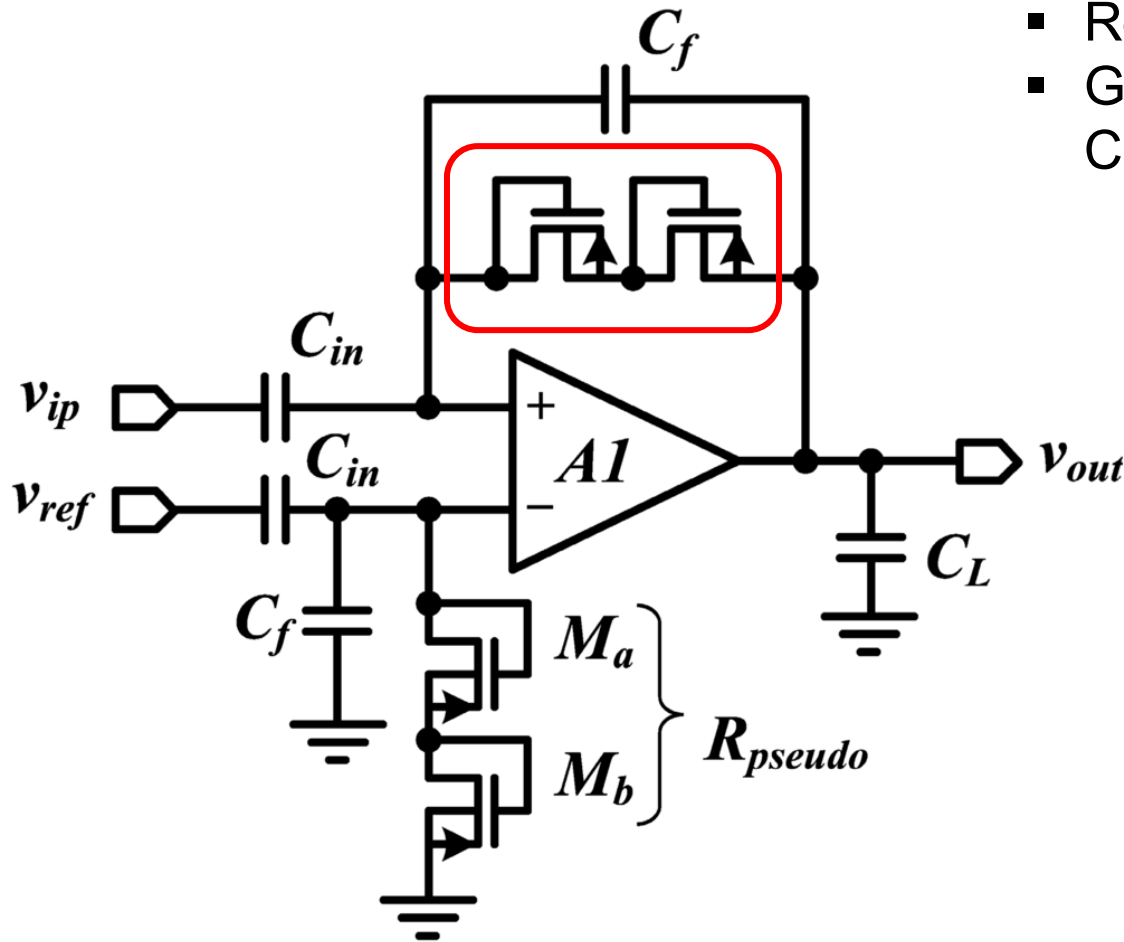




Voltage Sensing

Capacitively-coupled

- Achieve high Z_{in} without the need on input buffers
- Reject electrode offset
- Good matching between caps $\rightarrow$ High CMRR

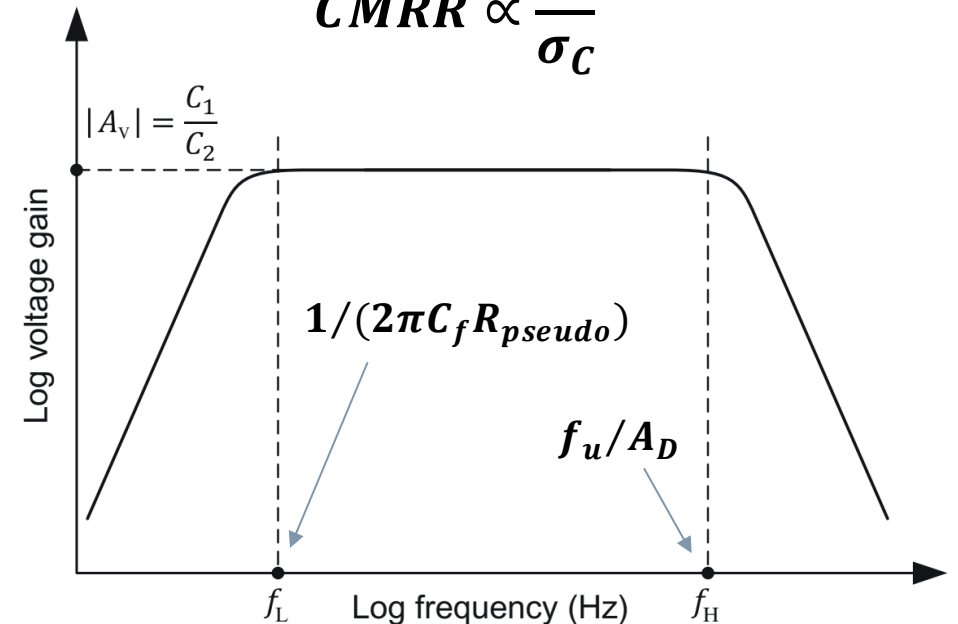


Most popular INA for ASIC implementation!

$$Z_{IN} = 1/j\omega C_{IN}$$

$$A_V = C_{IN}/C_f$$

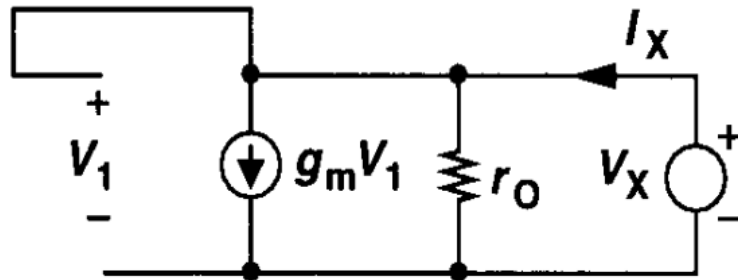
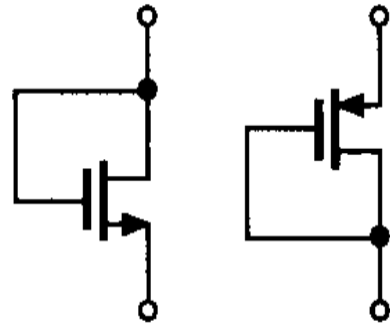
$$CMRR \propto \frac{1}{\sigma_C}$$



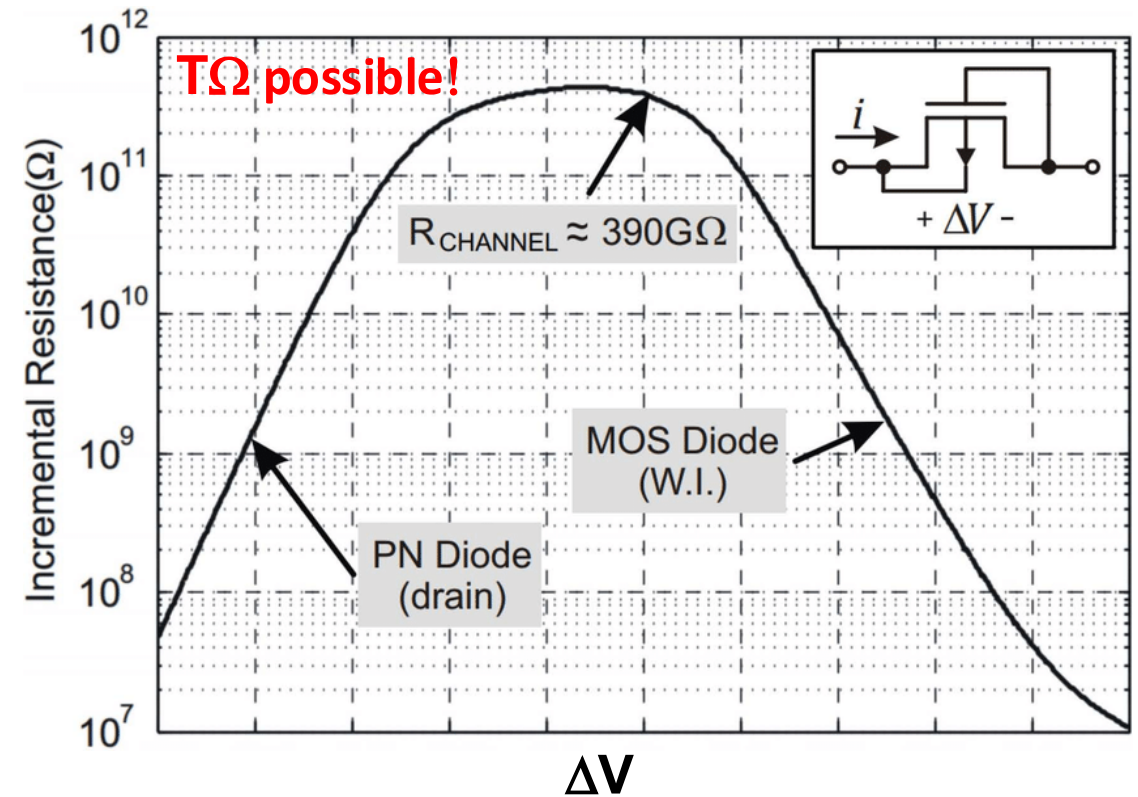
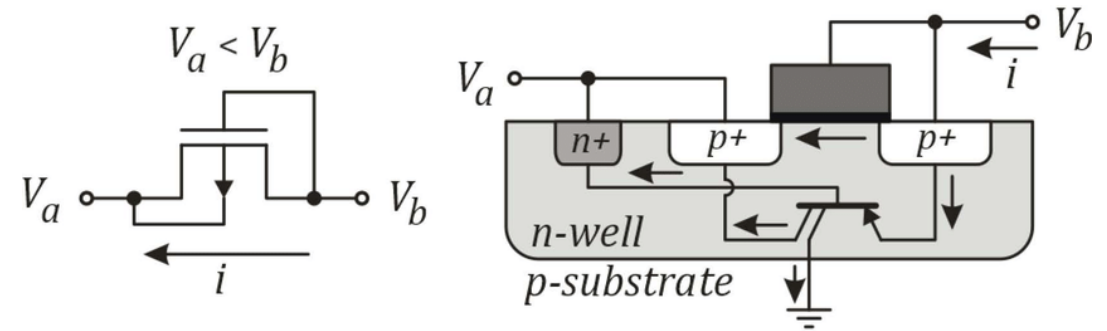


Pseudoresistor

Diode-connected MOS



$$R_X = (1/g_m) || r_o \approx 1/g_m$$

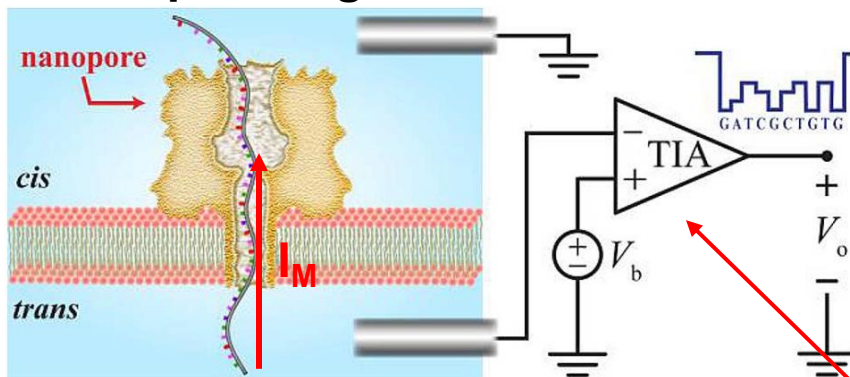


- Ha TMBE 2014
- Ch 3. Razavi

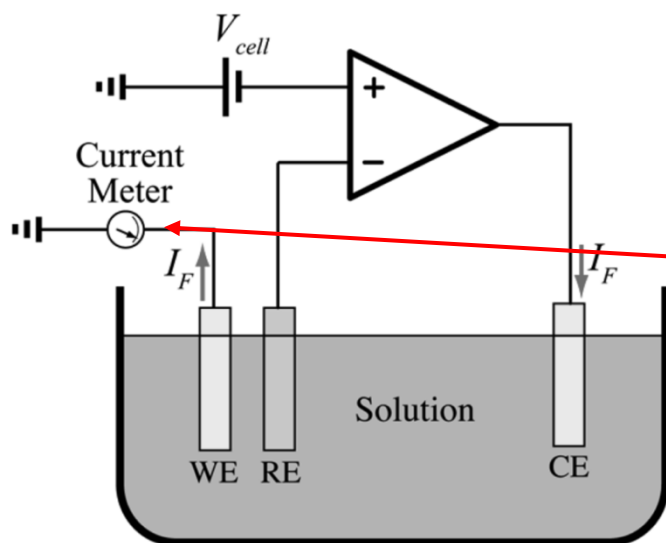


Current Sensing

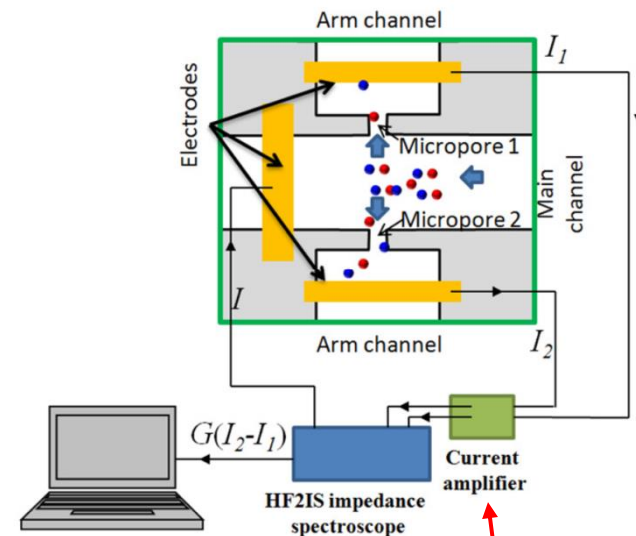
DNA Sequencing



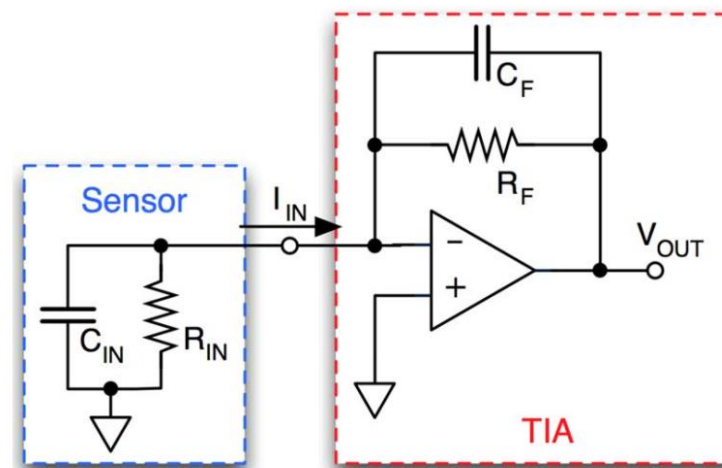
Electrochemistry - Potentiostat



Impedance spectroscopy

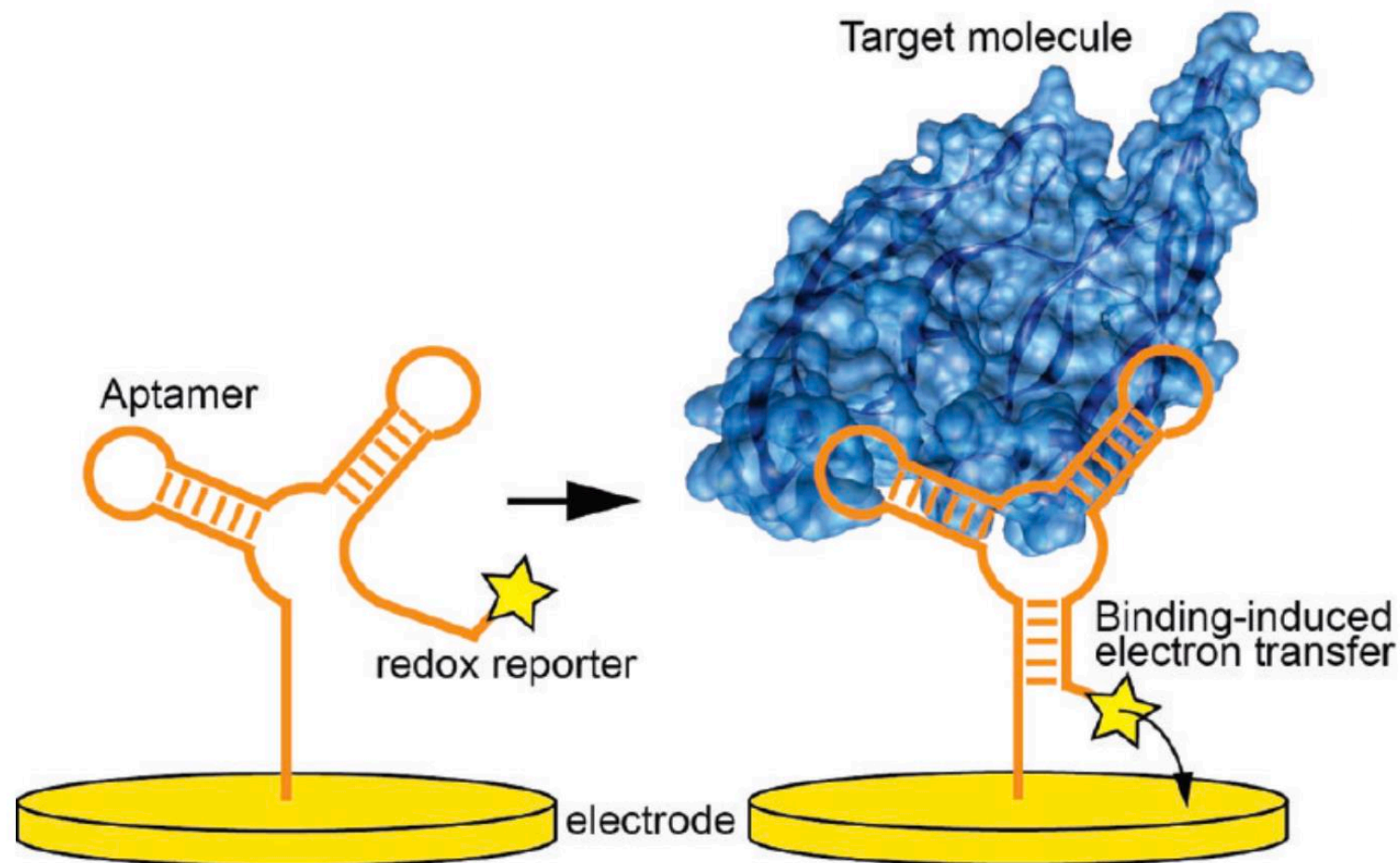


Transimpedance amplifier





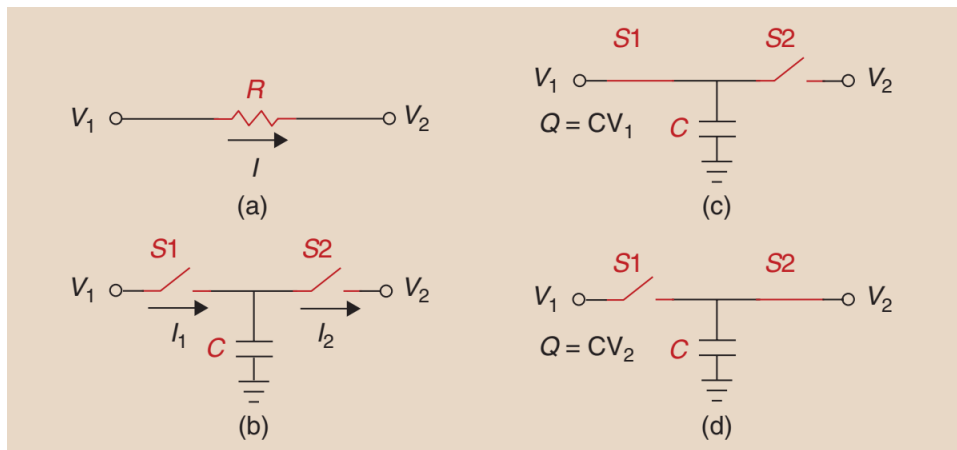
Charge-based Capacitive Sensing





Charge-based Capacitive Sensing

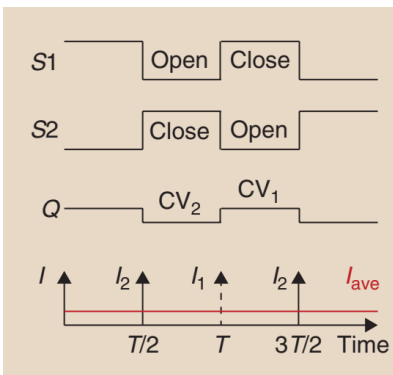
Switched-capacitor as a resistor



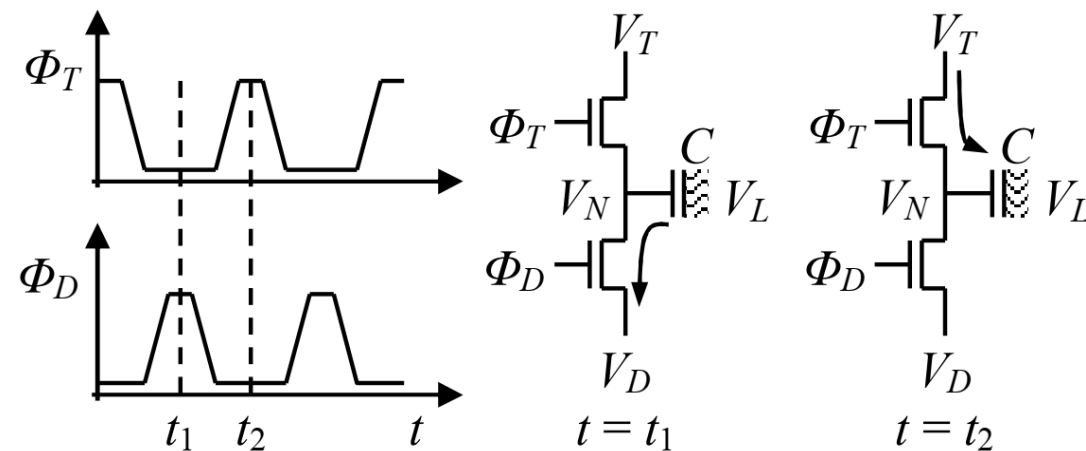
$$I_R = \frac{(V_2 - V_1)}{R}$$

$$I_{avg} = \frac{C(V_2 - V_1)}{T}$$

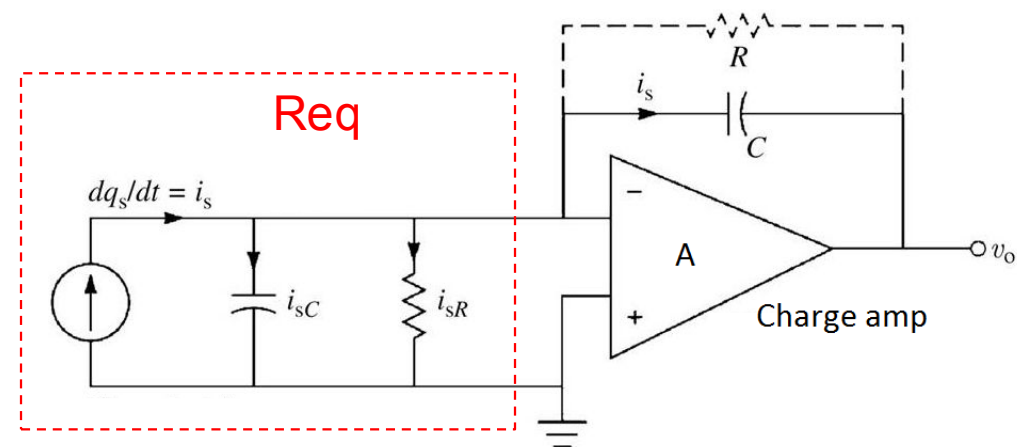
$$R_{eq} = \frac{T}{C}$$



Switched-cap sensing



$$\Delta I_S = \frac{C(V_T - V_D)}{T} = \frac{Q_S}{T}$$



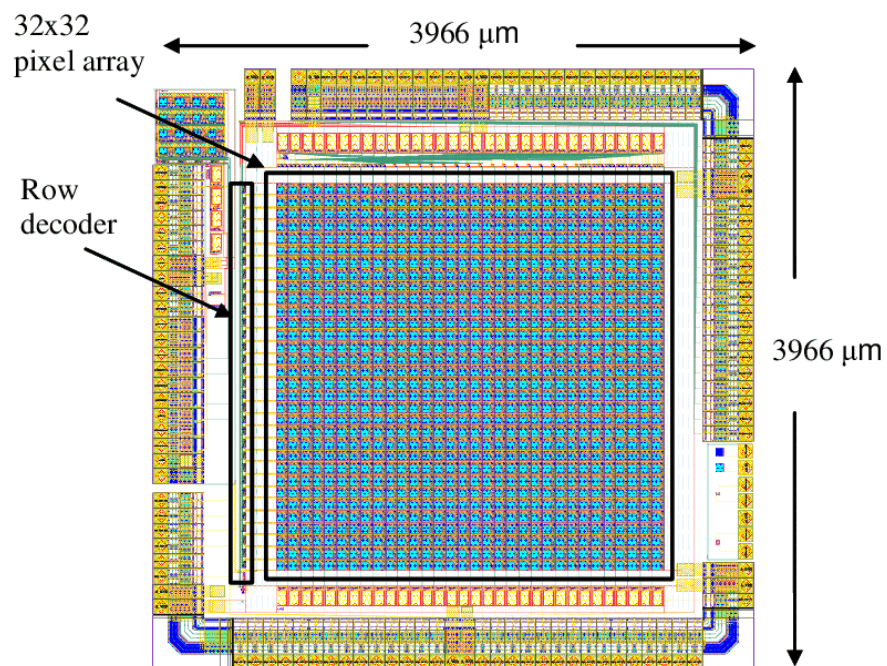
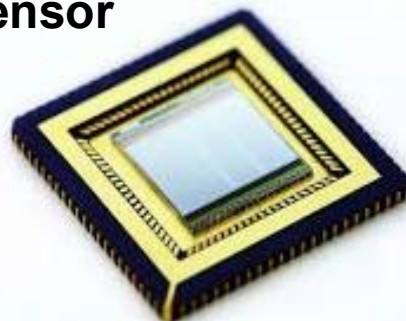


All about pixels (almost)



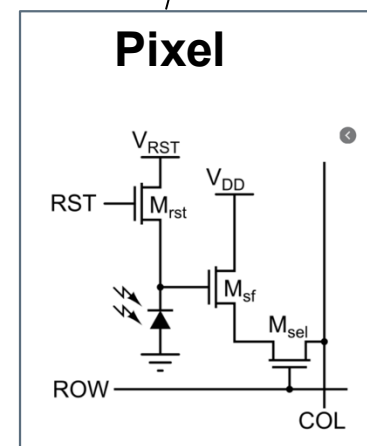
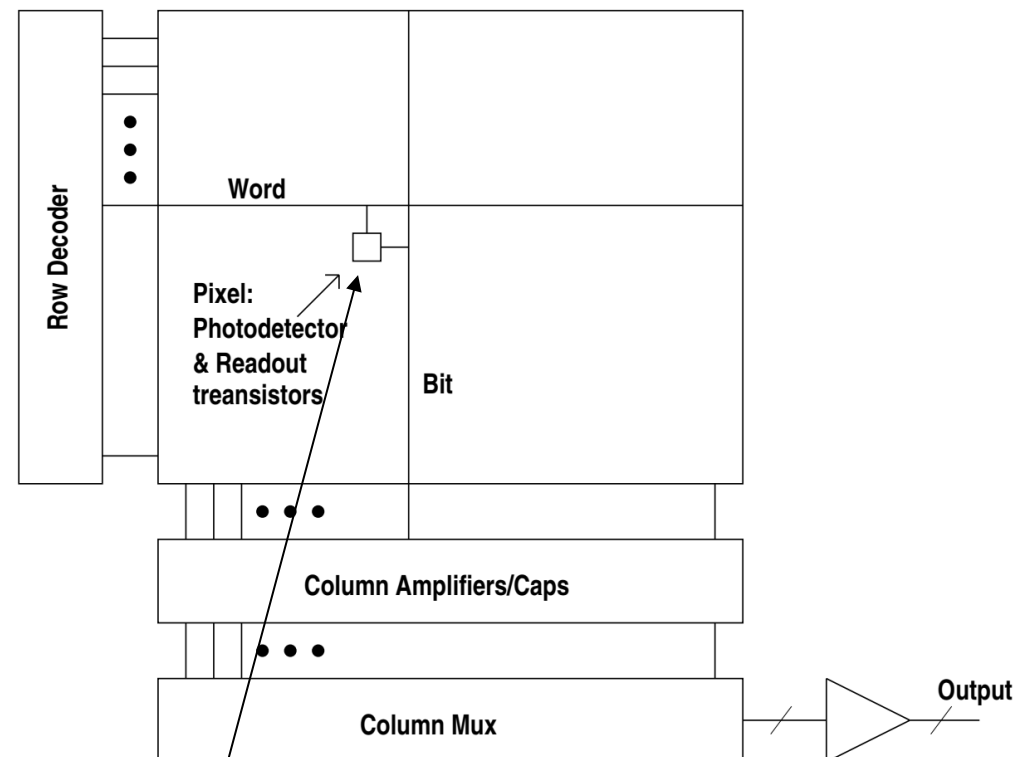
Pixel Array Chips

CMOS sensor chip

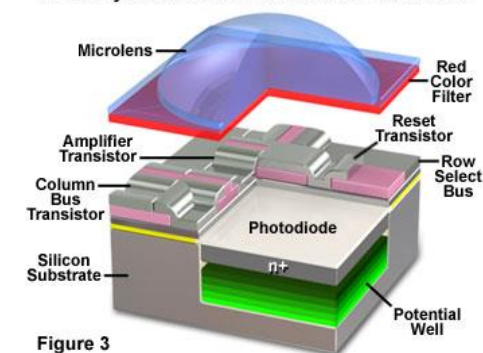


Layout

Architecture



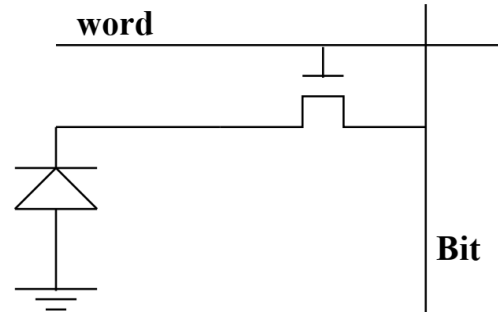
Anatomy of the Active Pixel Sensor Photodiode



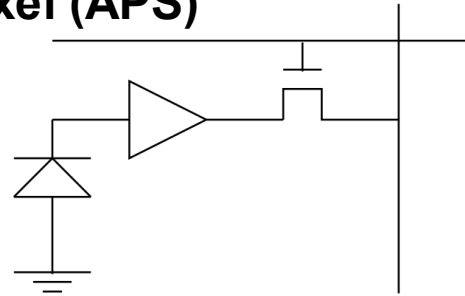


Pixel architectures

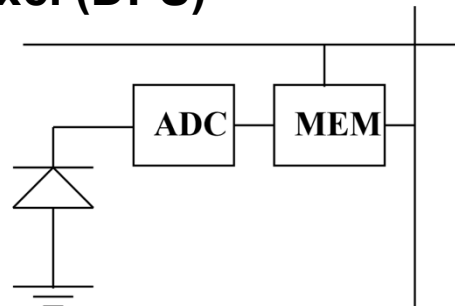
Passive pixel (PPS)



Active pixel (APS)



Digital pixel (DPS)



- **Passive pixel (PPS)**

- 1 T per pixel
- Small pixel, large fill factor
- Slow, low SNR

- **Active pixel (APS)**

- $>2T$ per pixel
- Faster, higher SNR
- Larger pixel, lower fill factor
- Most used

- **Digital pixel (DPS)**

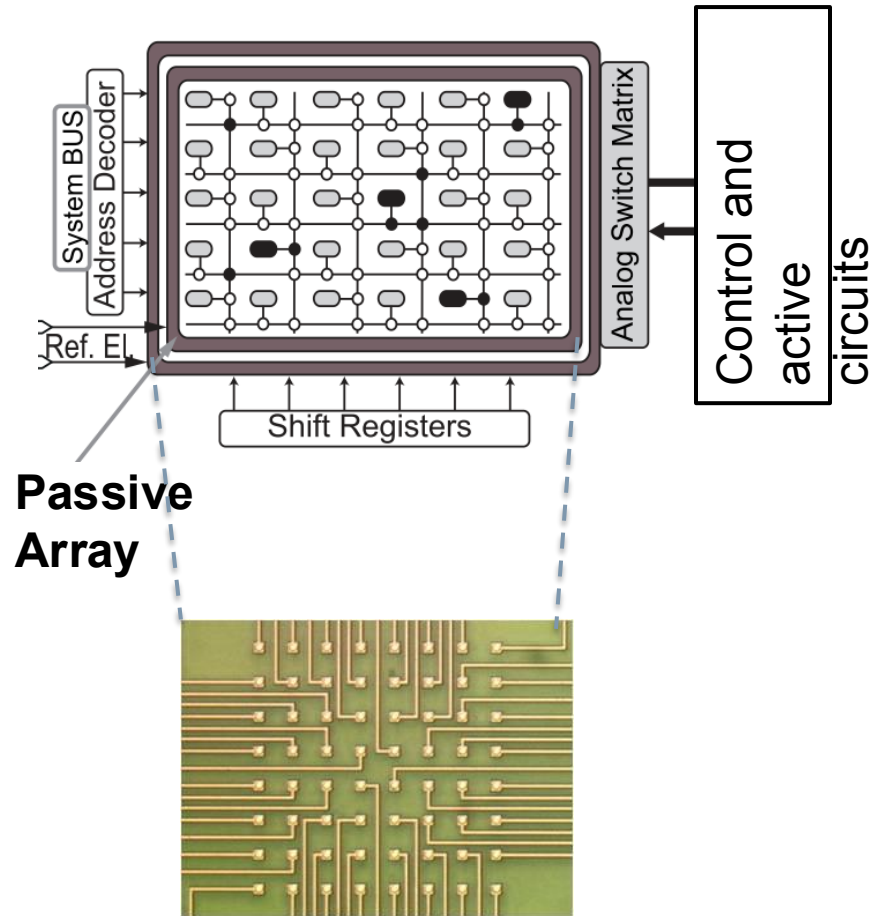
- $>5T$ per pixel
- Fast, digital signal
- Larger pixel, complex

This
lecture



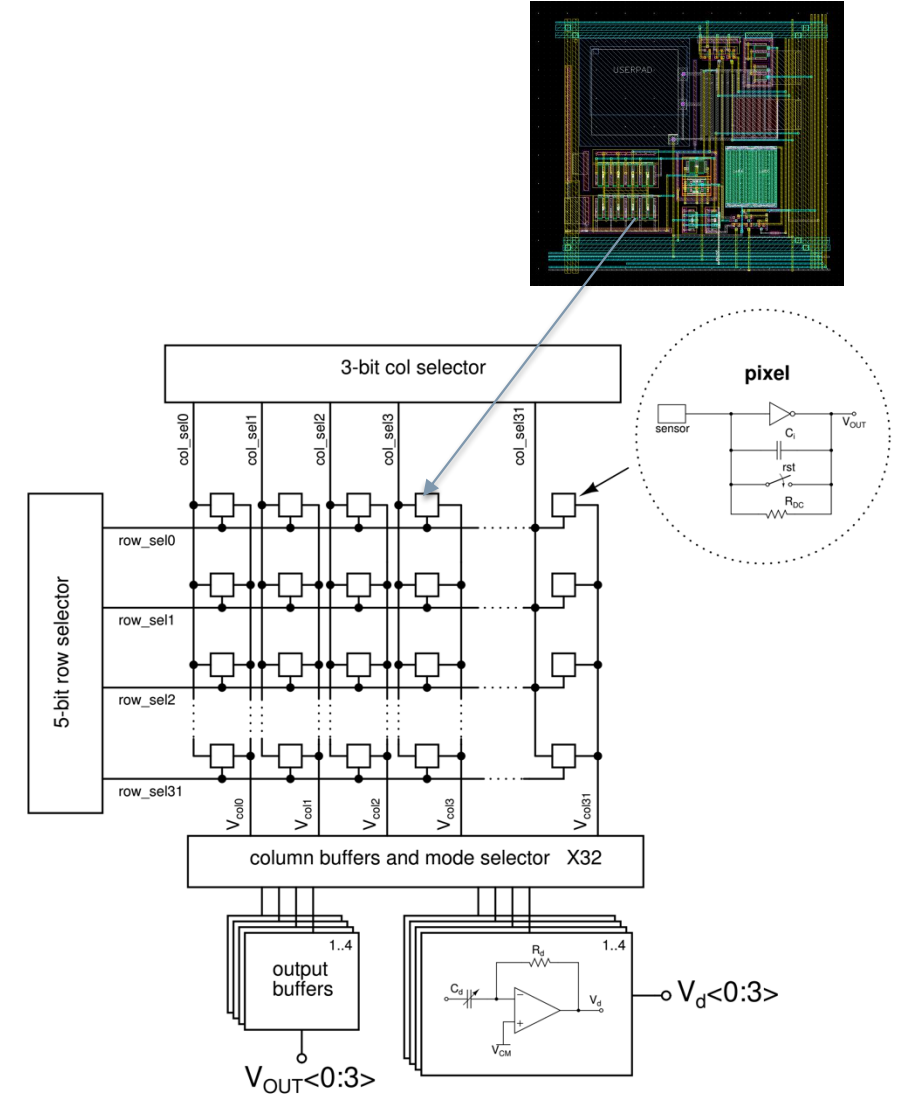
Passive Pixel (Switch matrix)

- Readout circuit **off-pixel**



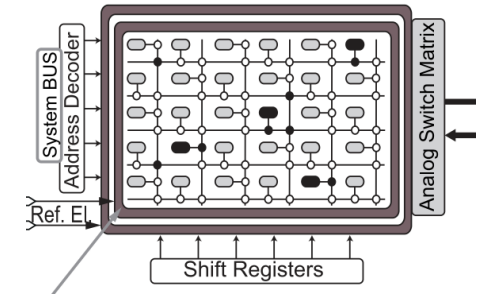
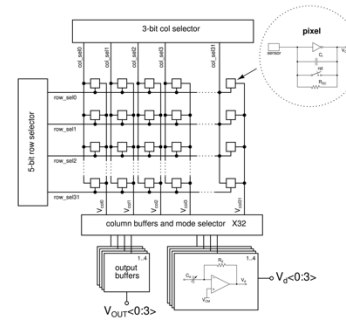
Active Pixel

Pixel layout





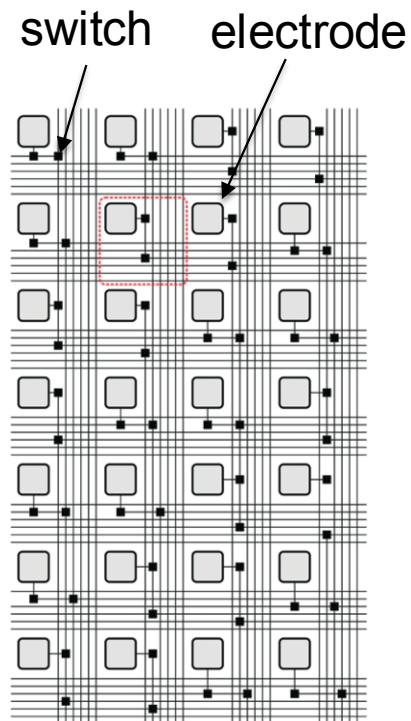
Active Pixels vs Switch Matrix



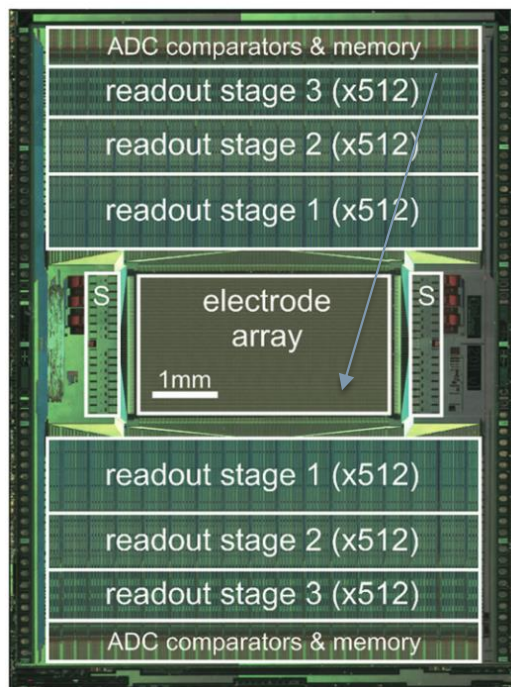
Specification	Units	APS	SM
Electrode density	El/mm ²	✗	✓
Array fill ratio	%	✗	✓
Spatial resolution	ppi	✗	✓
Frame rate	F/s	✓	✗
Power consumption	mW	✓	✗
Noise	μV_{rms}	0	0
SNR	dB	✓	✗
Sensing modalities	#	✗	✓



Passive Pixel CMOS MEAs



Switch Matrix



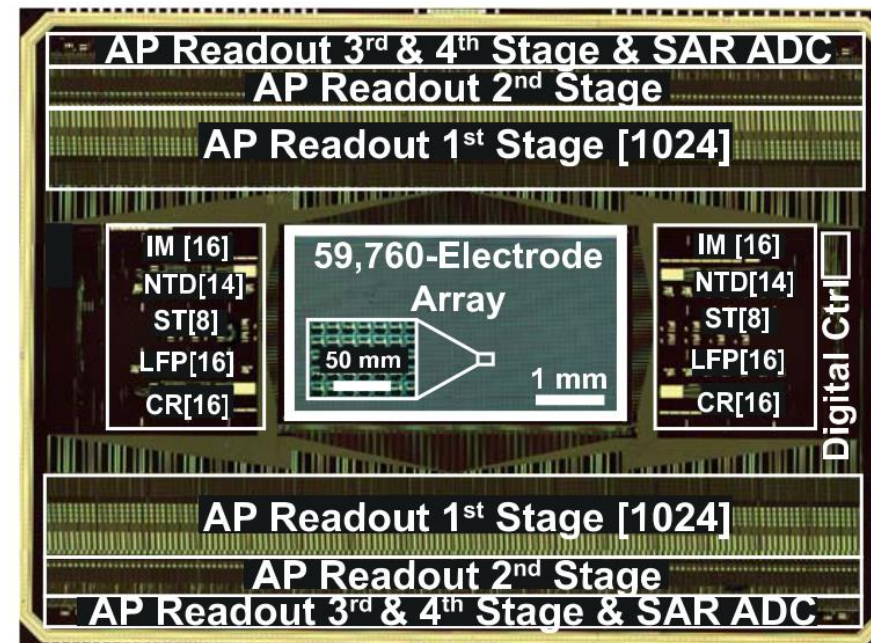
Muller Lab Chip 2015

Density: **3265 EI/mm²**

Noise: 12.4 μ Vrms (300Hz-10kHz)

Power: **75mW**

App: Neural recording



Dragas JSSC 2017

Density: **5487 EI/mm²**

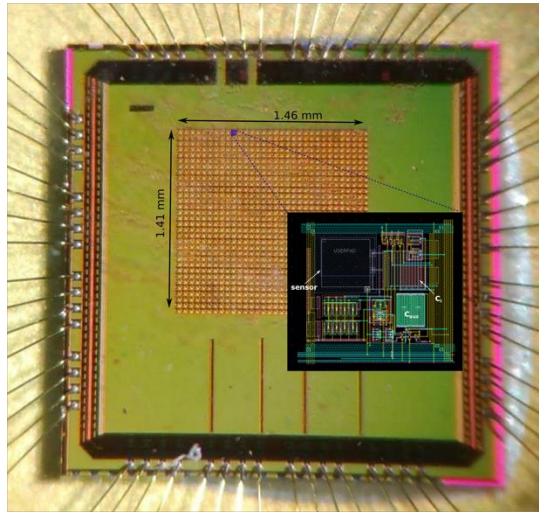
Noise: 12.4 μ Vrms (300Hz-10kHz)

Power: **86mW**

App: Neural recording

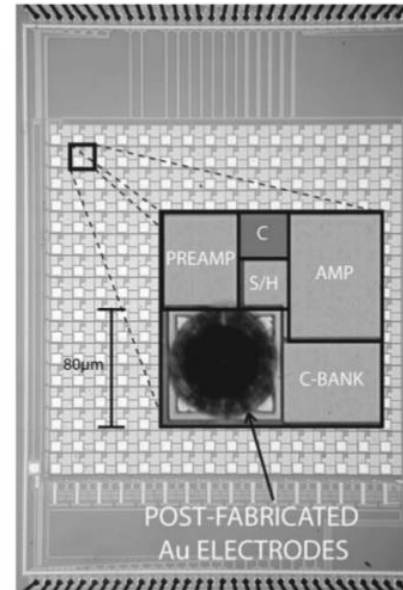


Active Pixels CMOS MEAs



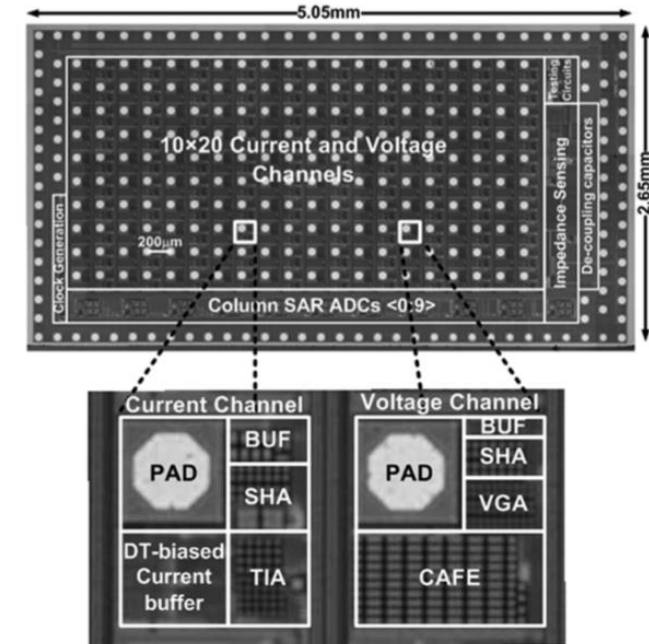
Valente ISCAS 2017

Electrodes: 1024
512 pxl/mm²
App: Single-cell
analytics



Aziz TBCAS 2007

Electrodes: **19 pxl/mm²**
Noise: 13 μVrms (10kHz)
Power: 6mW
App: Neural recording

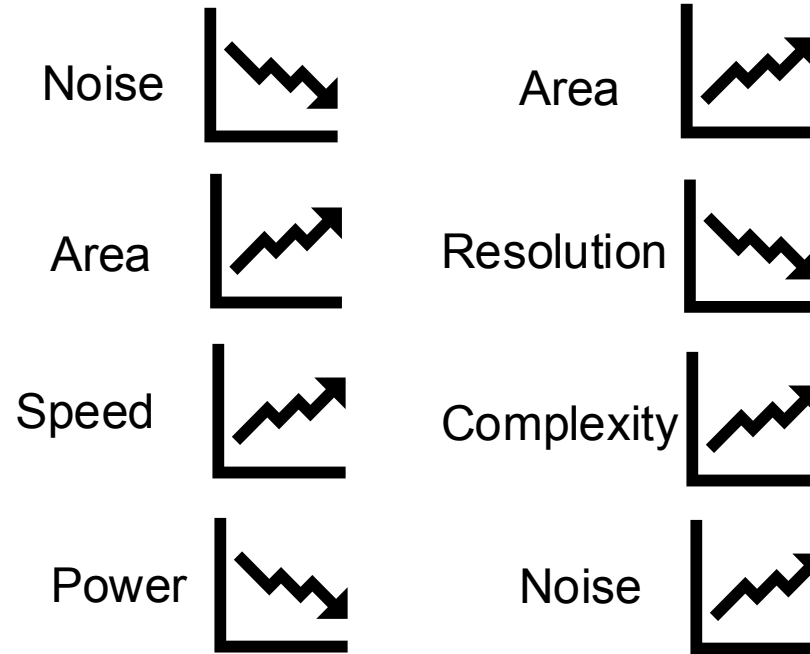
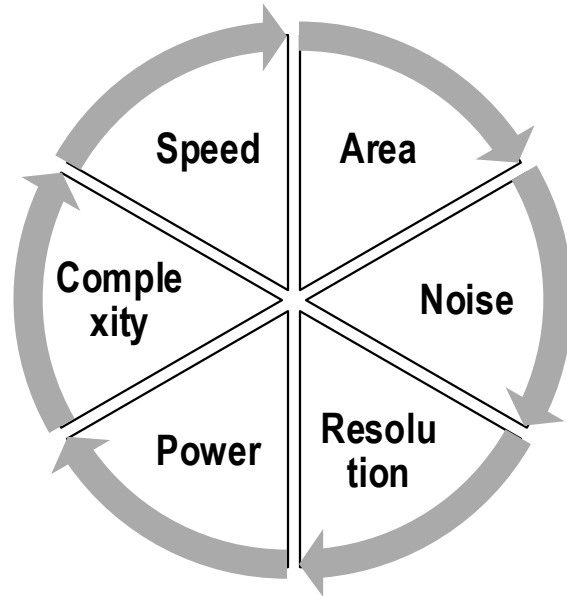


Guo TBCAS 2016

Electrodes: **16 pxl/mm²**
Noise: 4 μVrms
Power: 3.2mW
App: Neural recording



Pixel Design



Pixel architectures

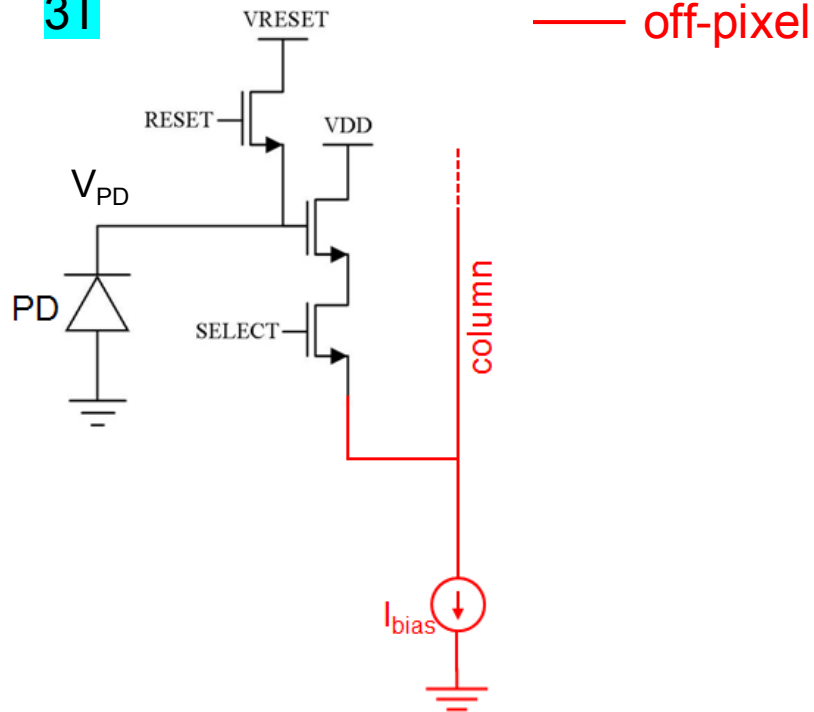
- Voltage-mode
- Current-mode
- Charge-based



Voltage Readout Circuits

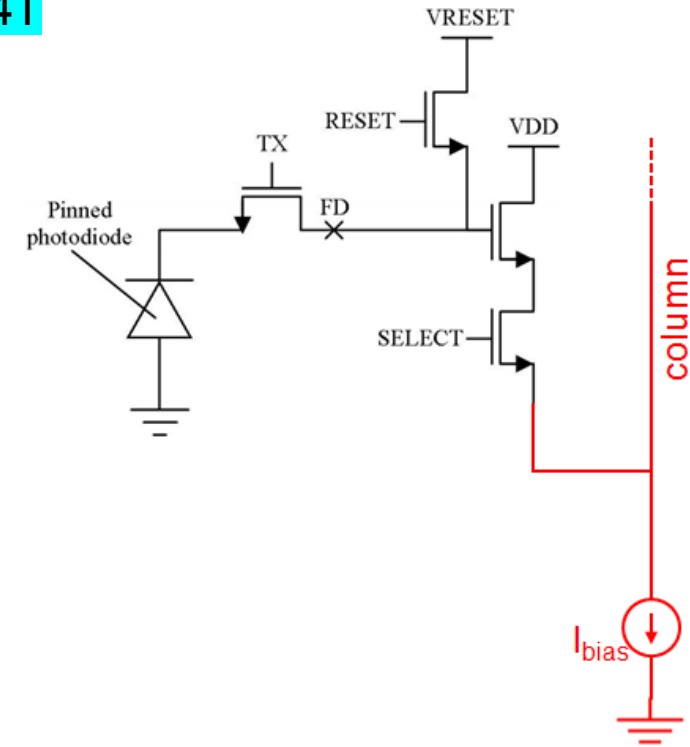
CMOS Imagers

3T



- Photodiode as voltage source
Can be a current source too!

4T



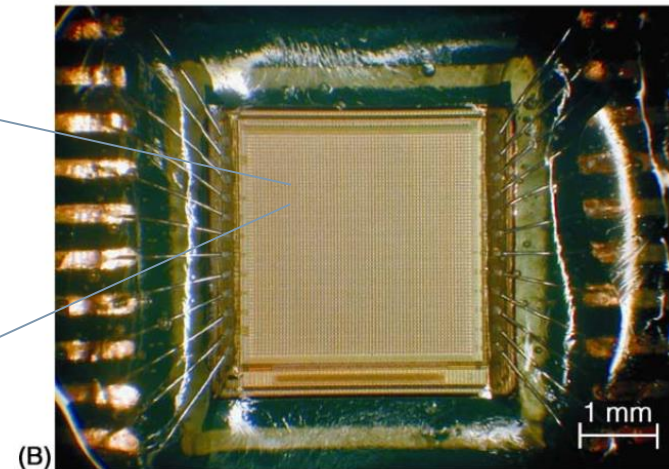
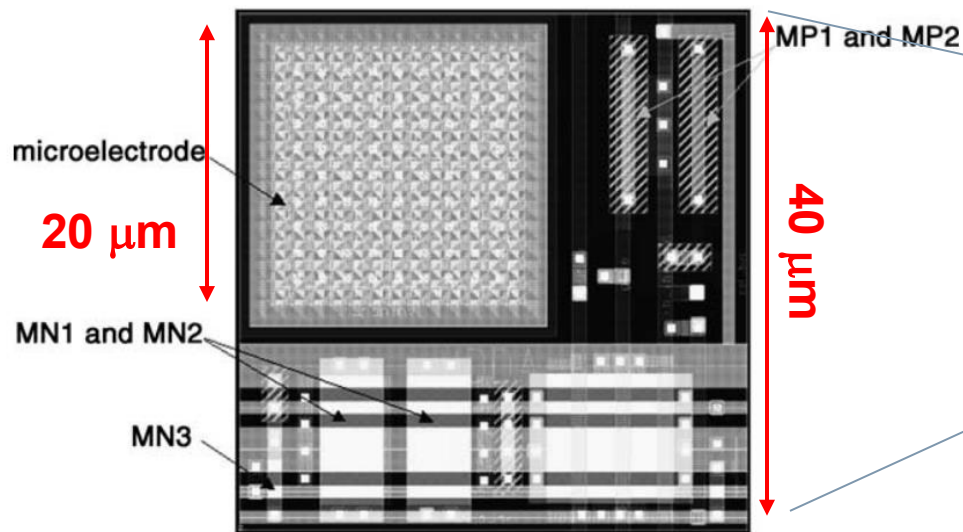
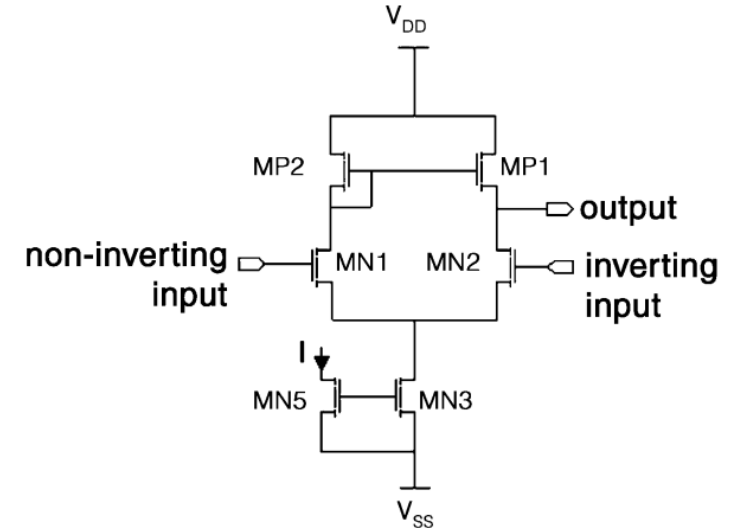
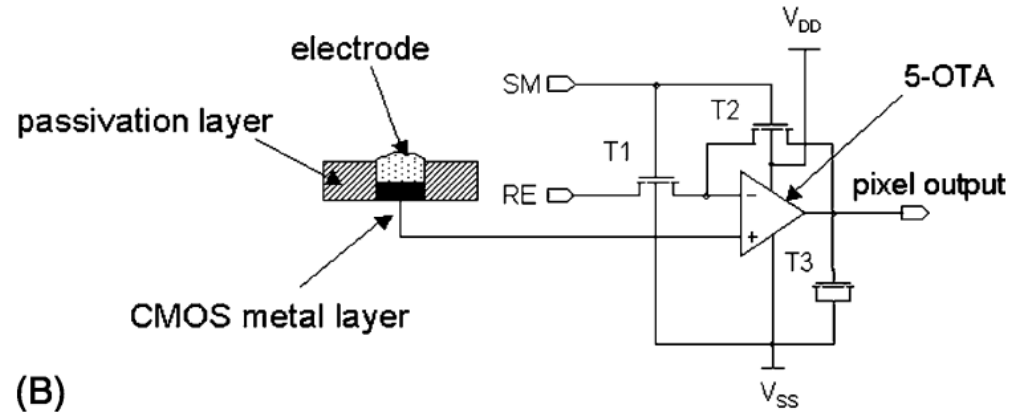
- 'Pinned' photodiode

What is the amplifier configuration in these active pixels?



Voltage Readout Circuits

MEA for neural recording



4096 pixels in 6.25 mm²
655 pixels/mm²

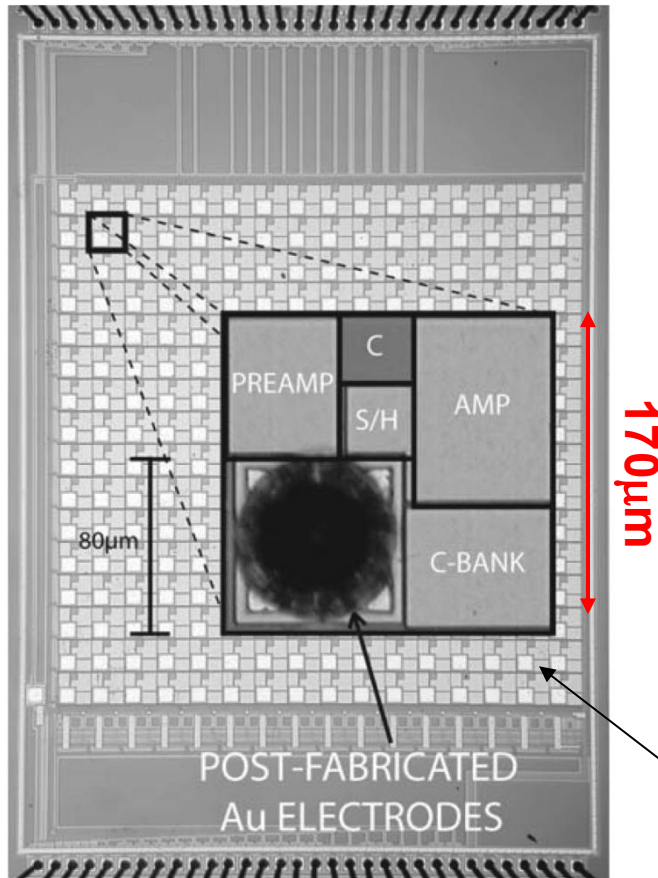
0.35μm CMOS

Berdondini BB '05

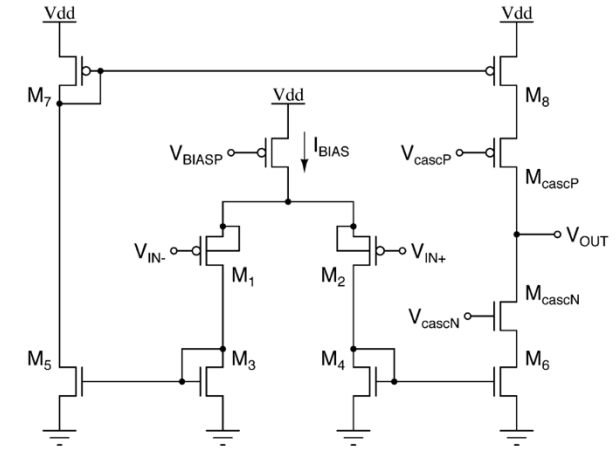
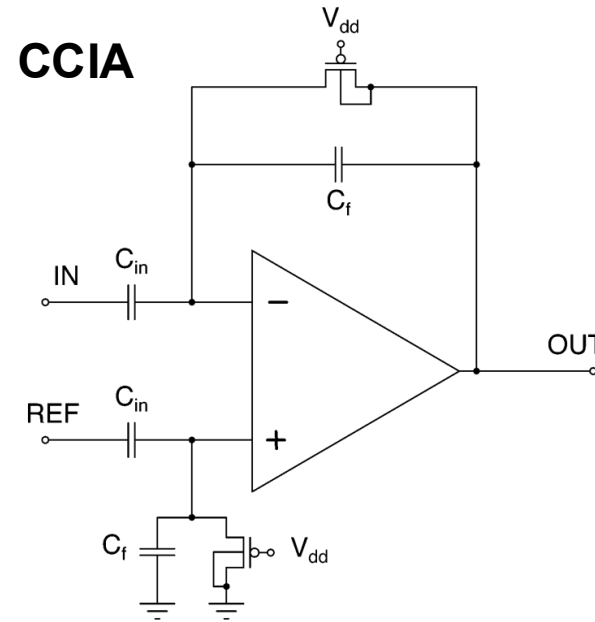


Voltage Readout Circuits

MEA for neural recording



256 pixels in 13.5mm²
19 pixels/mm²



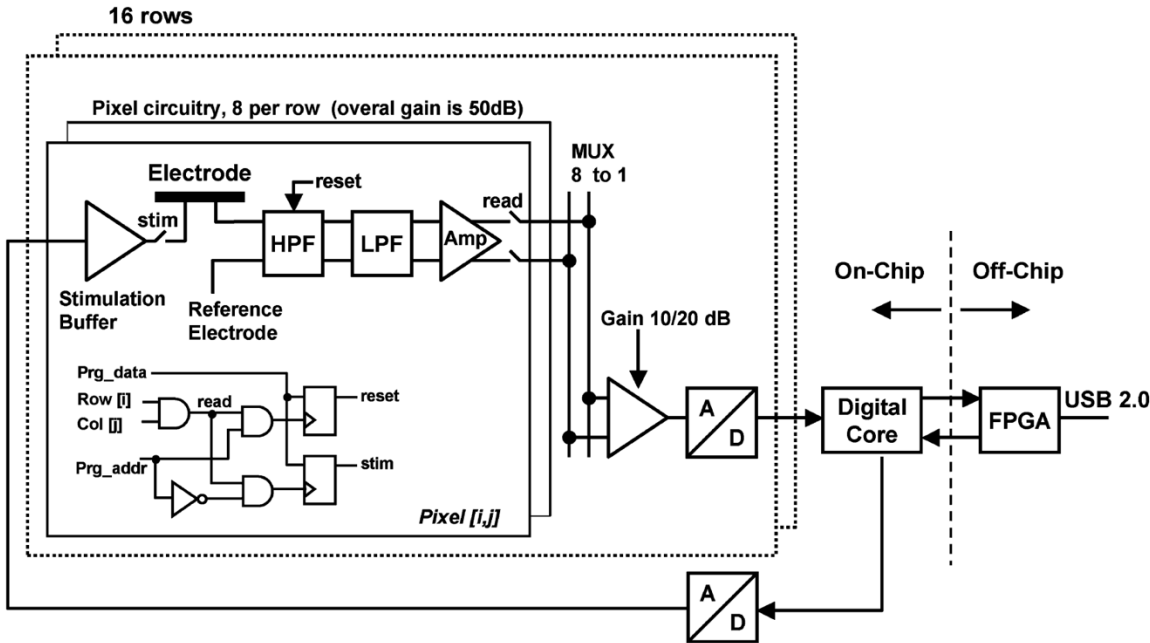
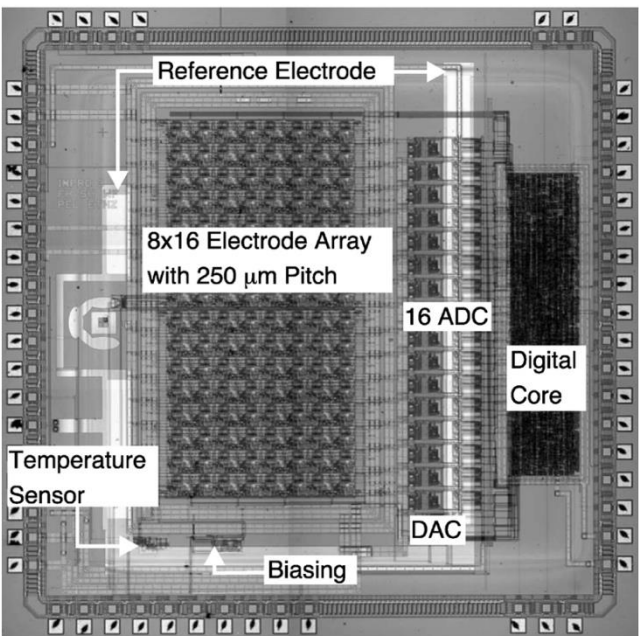
Channels	256
Max Sampling Rate	40kHz
Programmable Gain	200, 1000, 2500, 5000
Input-Referred RMS Voltage Noise	13µV over 10kHz
Output Voltage Range	1.5V
DC Offset Compensation	Double Sampling
LPF cut-off Frequency	1kHz-10kHz
HPF cut-off Frequency	<0.1Hz
Supply Voltage	3.3V
Core Power Dissipation	6mW
Technology	0.35µm, mixed-signal CMOS, double poly
Electrode Type	On-chip, Au; 100µm
Cell Pitch	170µm
Die Size	3×4.5 mm ²



Voltage Readout Circuits

MEA for neural recording

- 128 electrodes
- Sensing area: 8 mm²
- Resolution: 16 pixels/mm²



- Stimulation and sensing
- > 40T!!

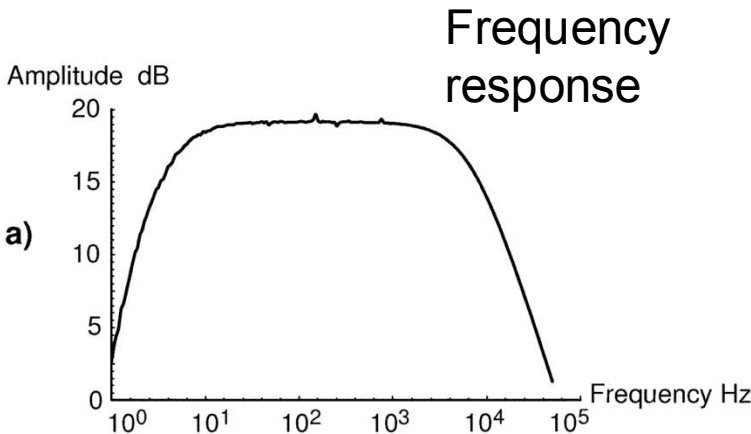


TABLE I
SUMMARY OF DEVICE CHARACTERISTICS

Parameter	Value
Supply Voltage	5 V
Power Consumption	120 mW
Chip Size	6.5x6.5 mm ²
Process	0.6 μm CMOS with 2P, 3M, HR
Number of Electrodes	128
Sensor Pitch	250 μm
Electrode Sampling Rate	20'000 1/s
Data Rate (In)	0.4 MB/s
Data Rate (Out)	3.2 MB/s
Sensor Area	2x4 mm ² , depends on post-processing
Amplification	1000 or 3000



Current Readout Circuits

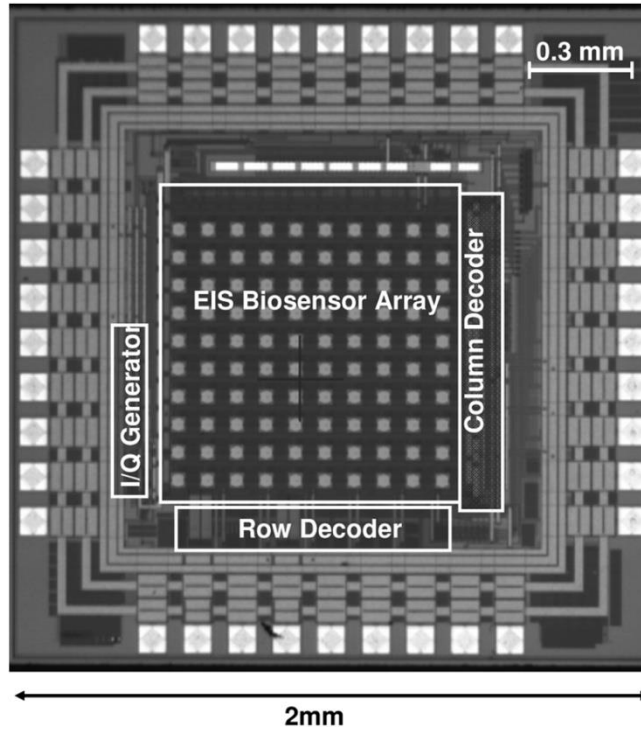
- Transimpedance Amplifier
- Current Mirror
- Switched-Current Circuits
- Current Conveyors
- Current Feedback Opamp



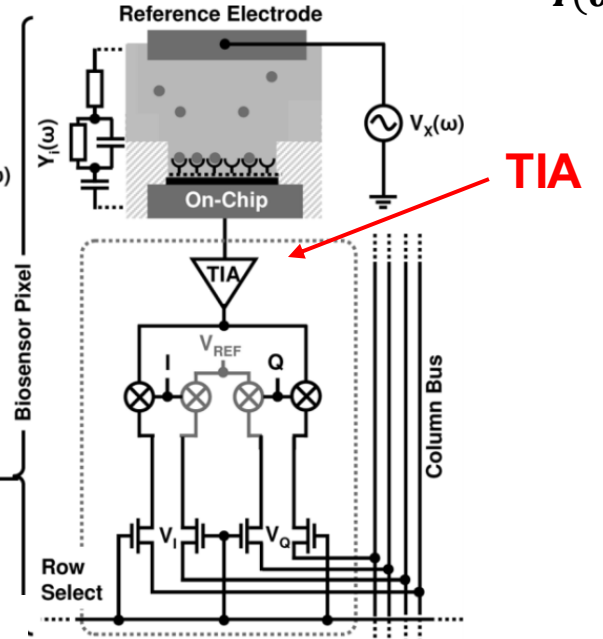
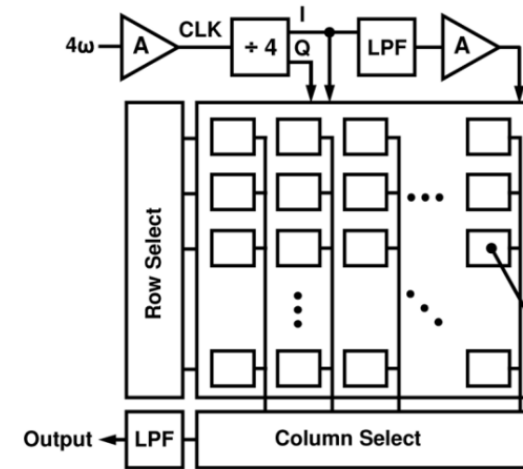
Current Readout Circuits – CT

Impedance measurement

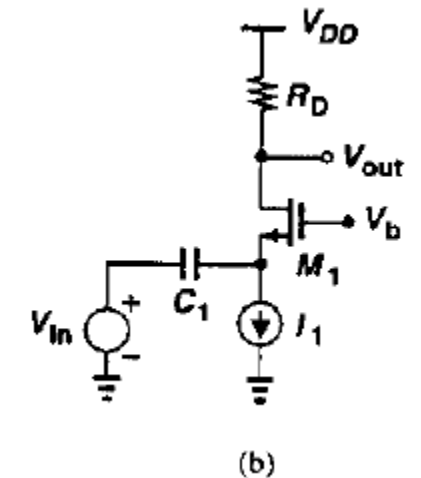
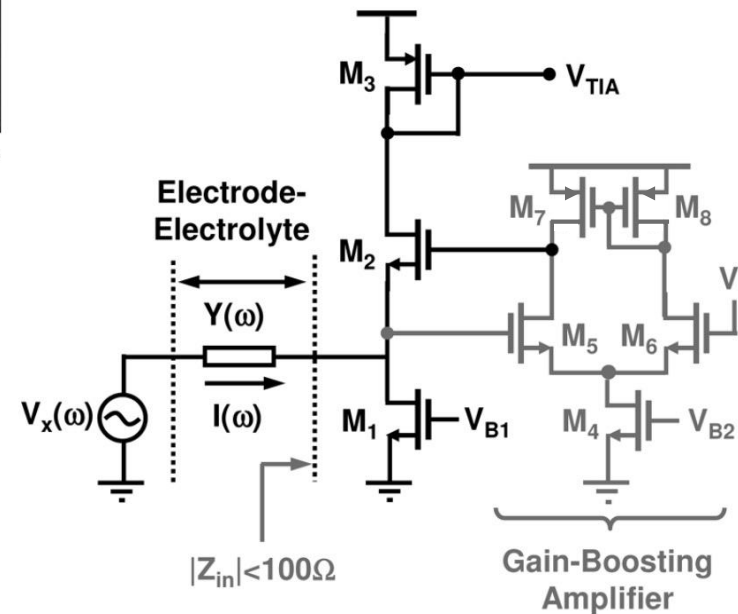
$$Z(\omega) = \frac{V(\omega)}{I(\omega)}$$



- 100 Electrodes
- 1 mm² array
- 100 EI/mm²



In-pixel analog signal processing





The diagram illustrates a current-mode integrator circuit. An operational amplifier (A) is configured with a feedback capacitor C_{fb} and a reset switch (RST). The input stage consists of a current source I_{pd} in parallel with a capacitor C_{pd} . The output voltage v_{out} is shown as a sawtooth waveform, indicating the integrator's function.

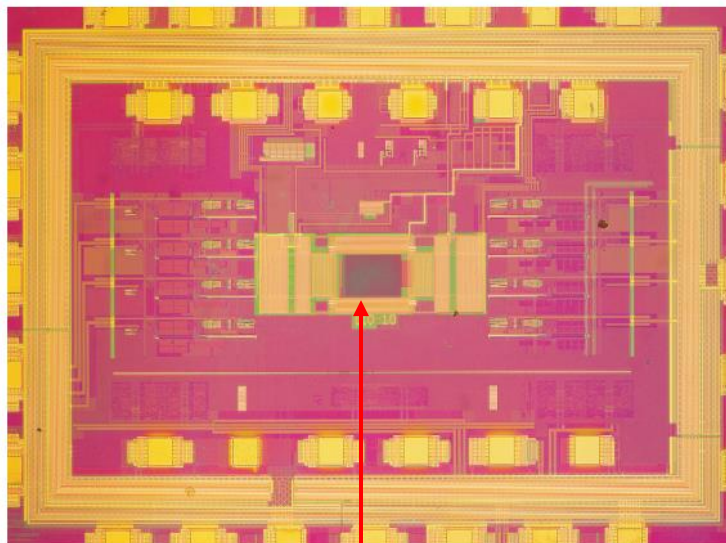
-
- The schematic diagram of the proposed column output circuit is shown in Figure 1. The circuit is a differential pair with a feedback loop. The feedback loop consists of a capacitor C_{fb} and a resistor RST . The cascode current source is formed by transistors $M4$ and $M3$. The differential pair consists of transistors $M1$ and $M2$. The output is taken from the node between $M3$ and $M4$, labeled RS . The circuit is connected to VDD and VSS , with biasing voltages V_{cp} and V_{bp} applied to the gates of $M6$ and $M7$ respectively. The output is connected to the To DDS circuit.



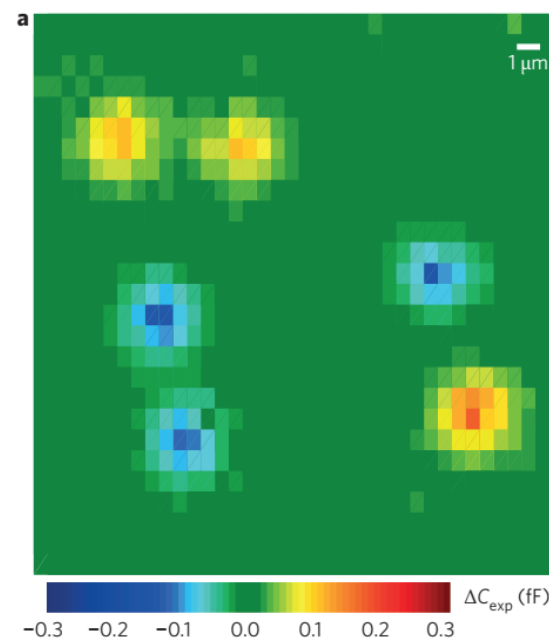
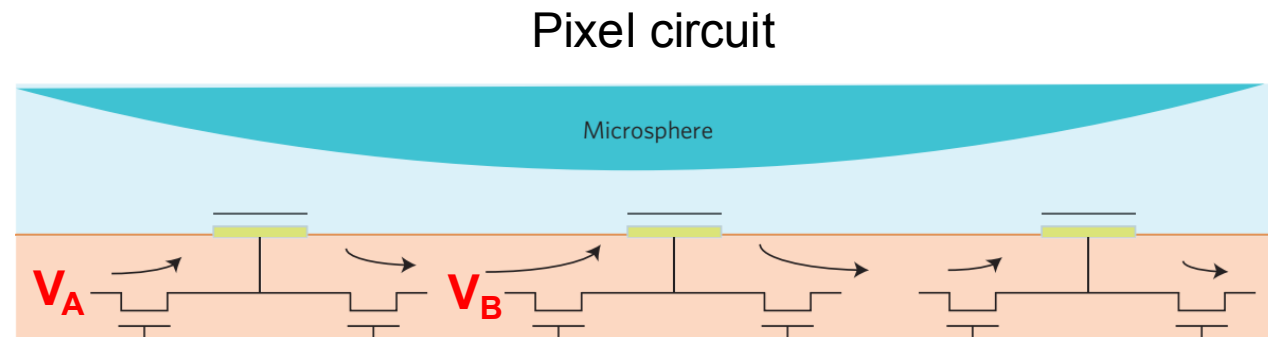
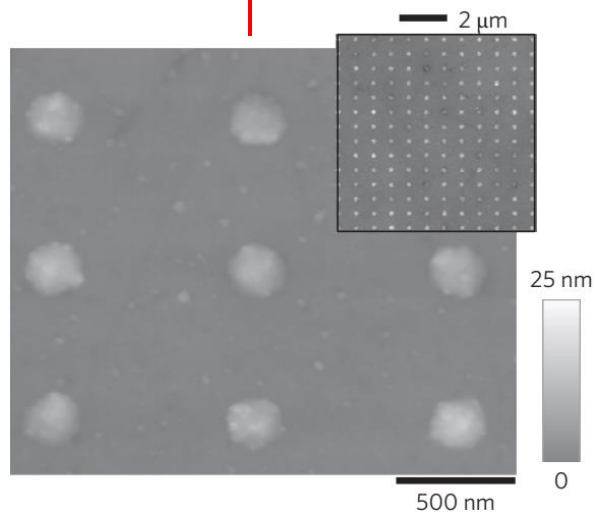


Charge Readout Circuits

Molecular sensing



Nanocapacitor chip



- Off-chip current measurement
- 65536 pixels!
- 0.6 μm X 0.89 μm

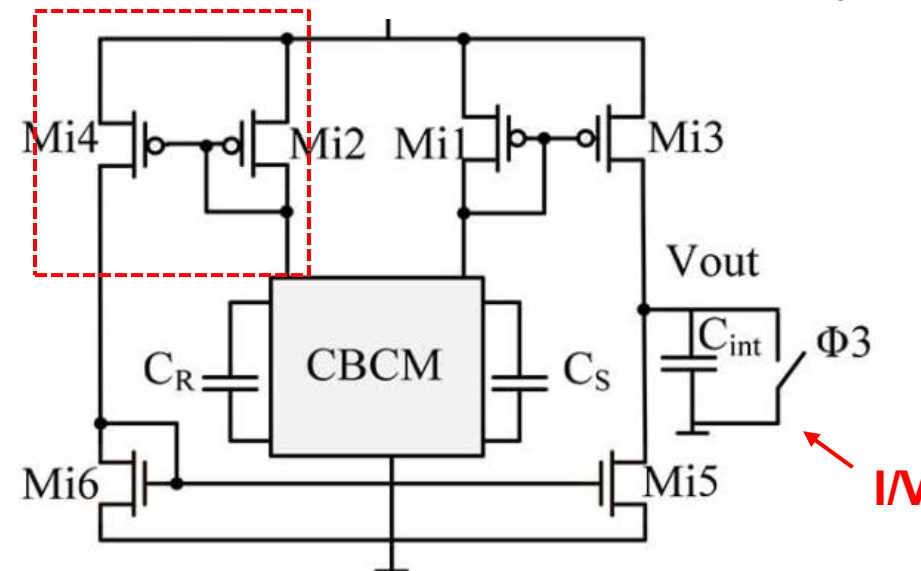
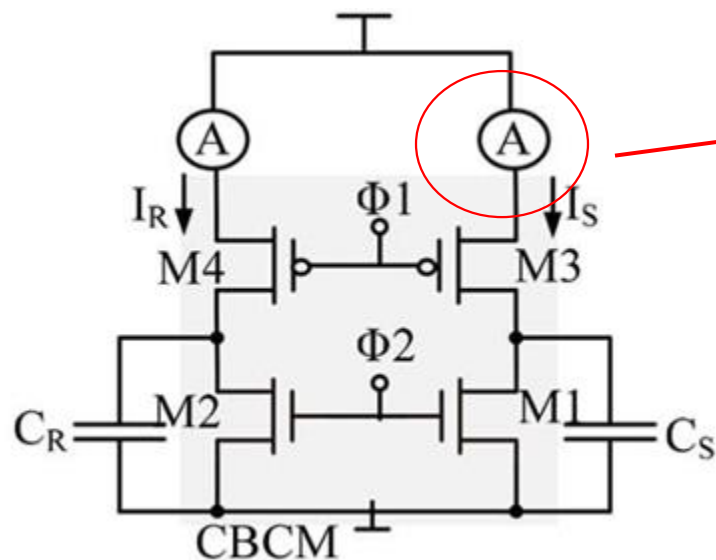
Laborde
nnano '15

Images based on cap change

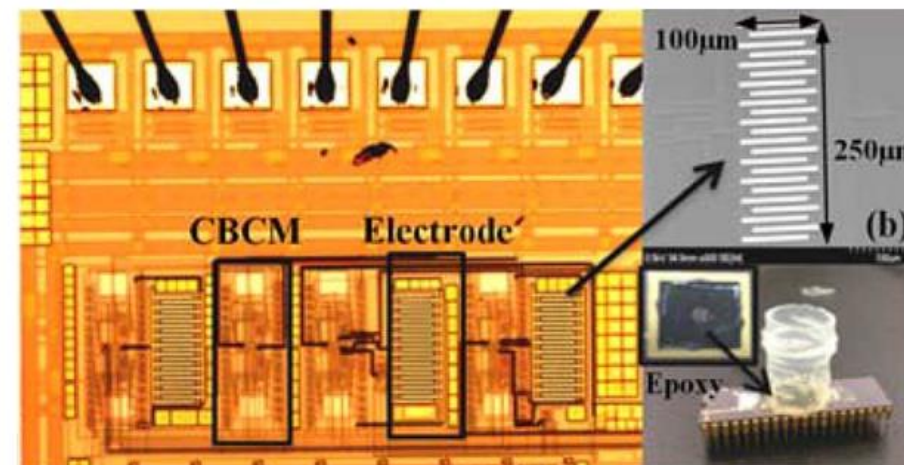
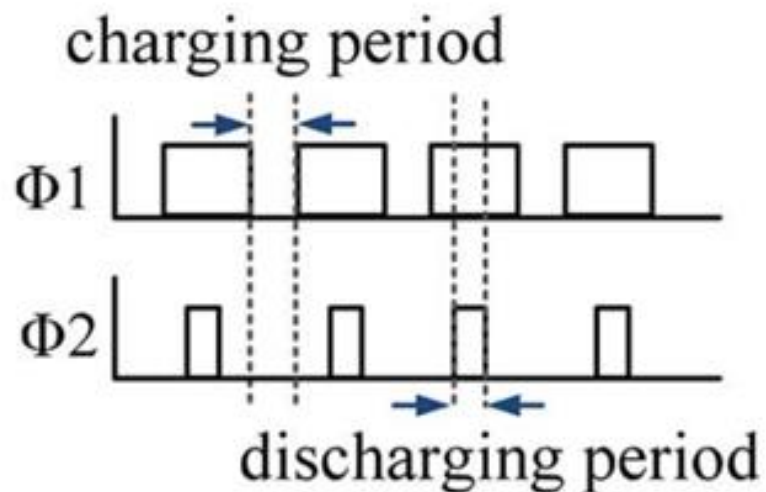


Charge Readout Circuits

Chemical sensing



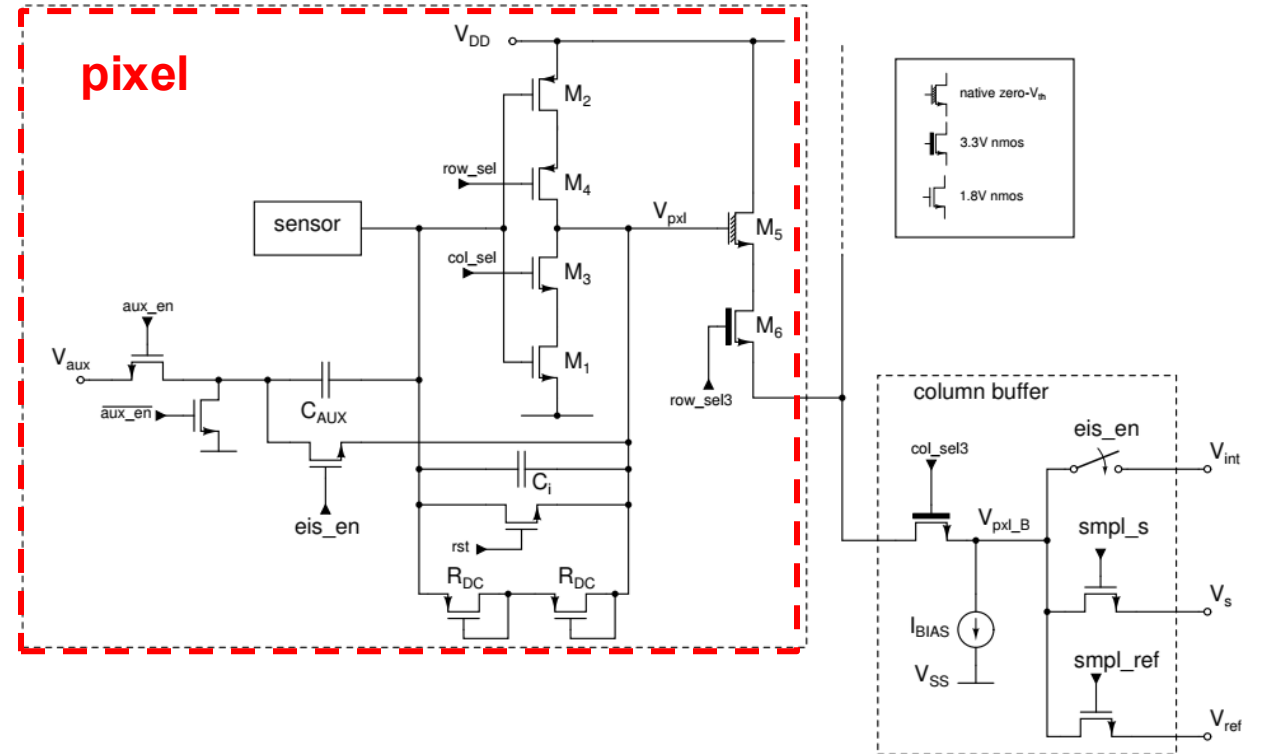
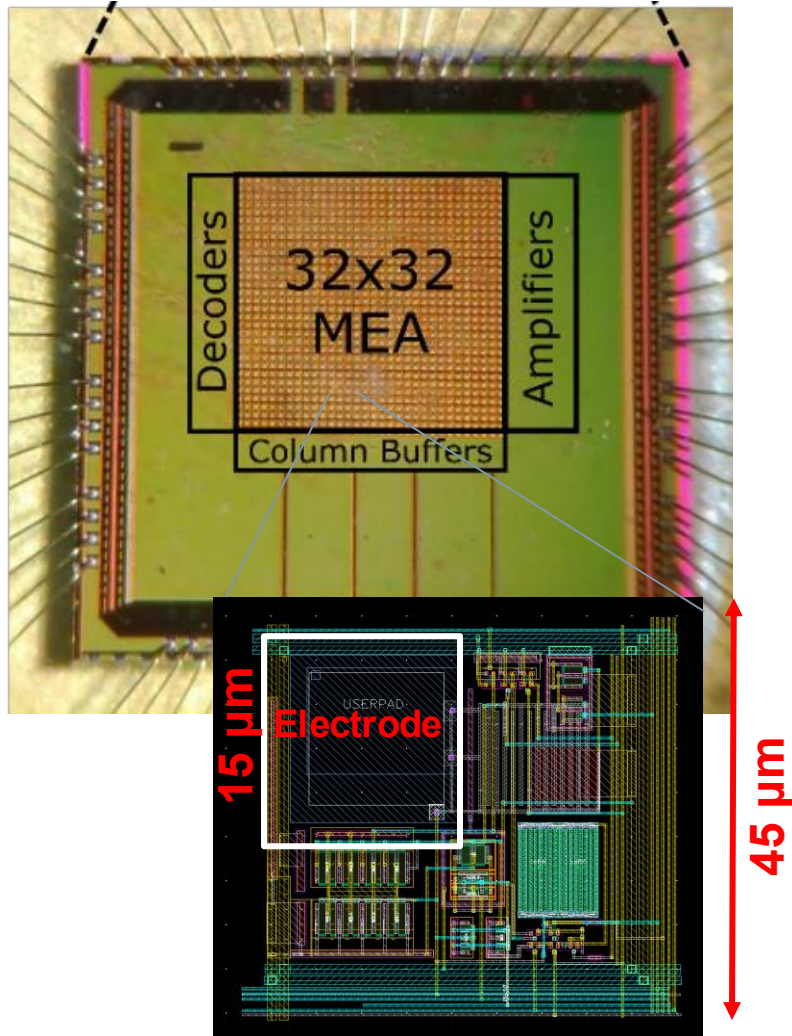
$$V_{out+} = \frac{K\Delta C(VDD) \cdot f_s}{C_{int} f_{int}}$$





Multimodal Readout Circuits

Charge and current readout



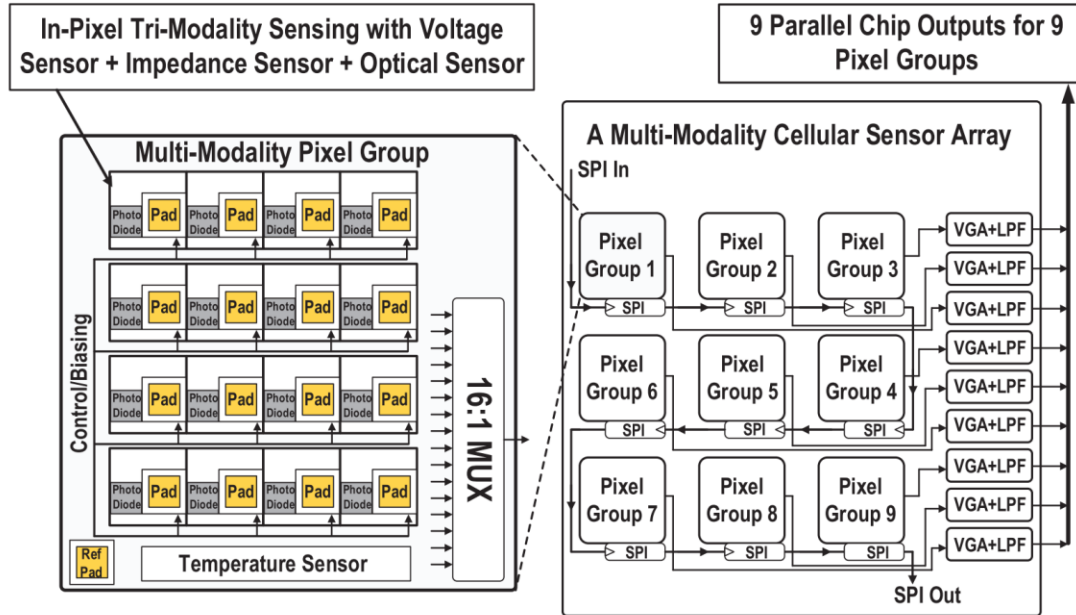
- 1024 pixels
- 2 mm²
- 512 EI/mm²

Valente ISCAS '17

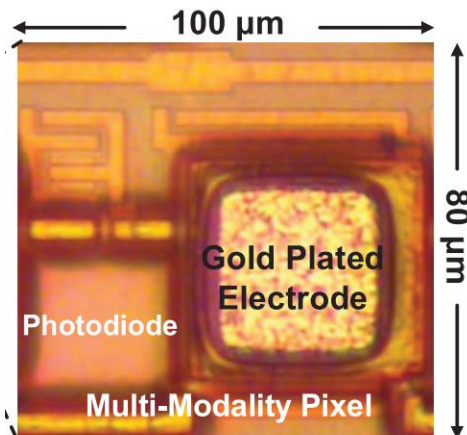


Multimodal Readout Circuits

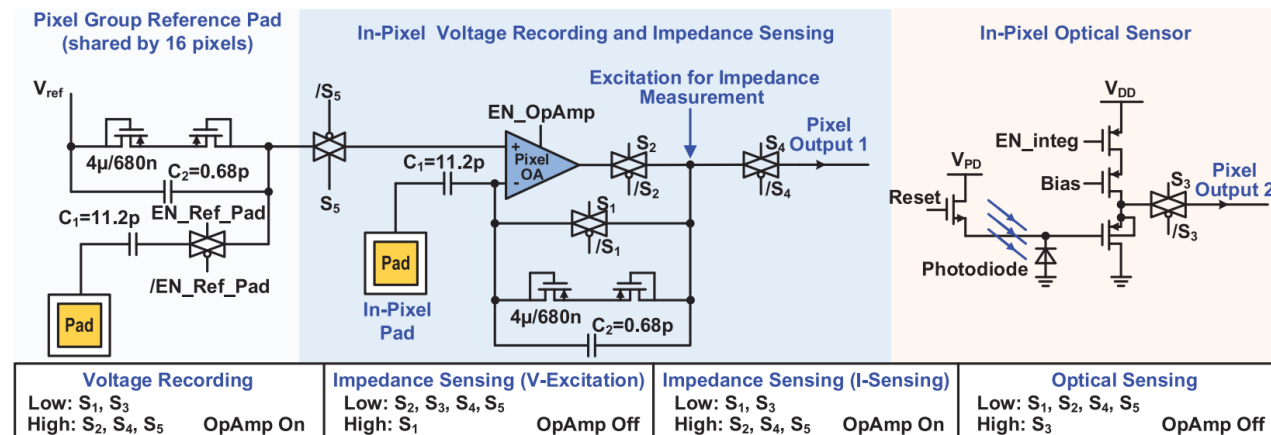
Voltage/current electrical & optical



- **PD:** $40\mu\text{m} \times 40\mu\text{m}$
- **EL:** $40\mu\text{m} \times 40\mu\text{m}$
- **Pixel:** $80\mu\text{m} \times 100\mu\text{m}$
- 144 pixels
- 2.25 mm^2 sensing area
- 64 pixels/mm^2



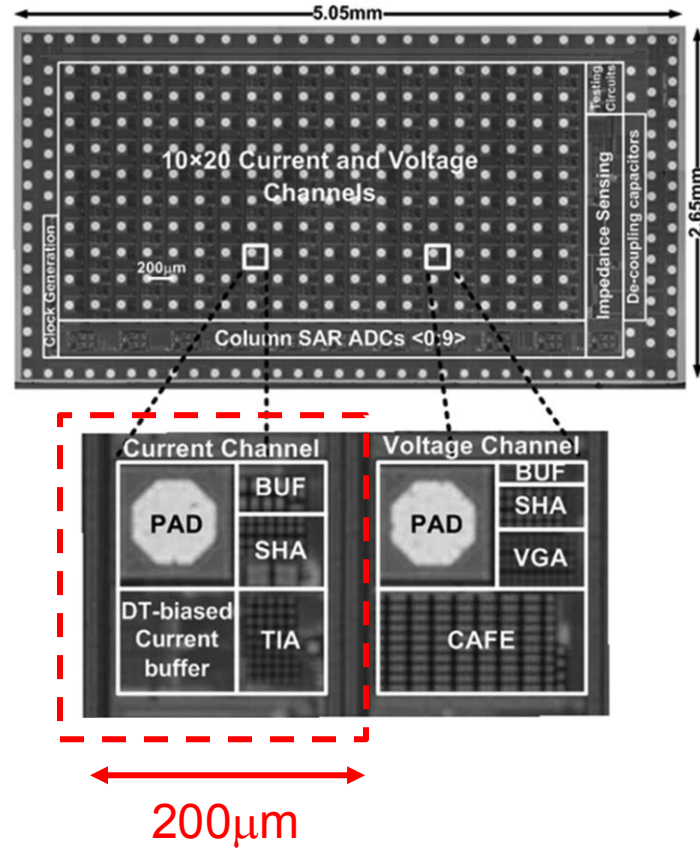
CCIA





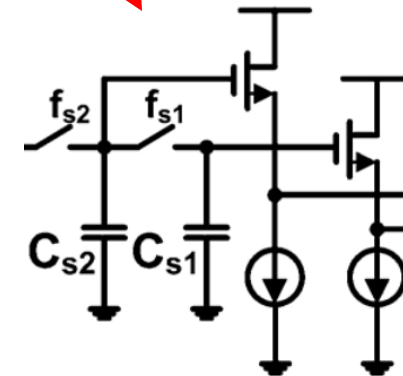
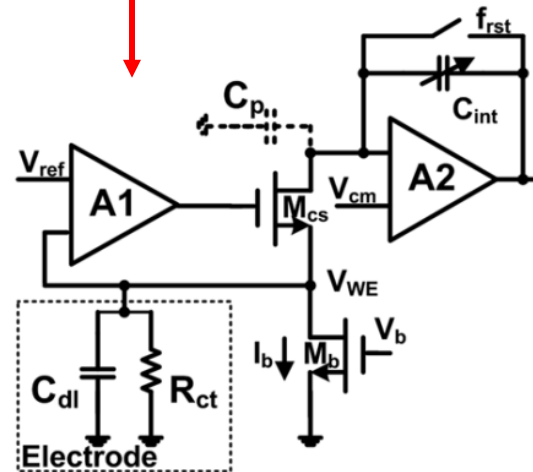
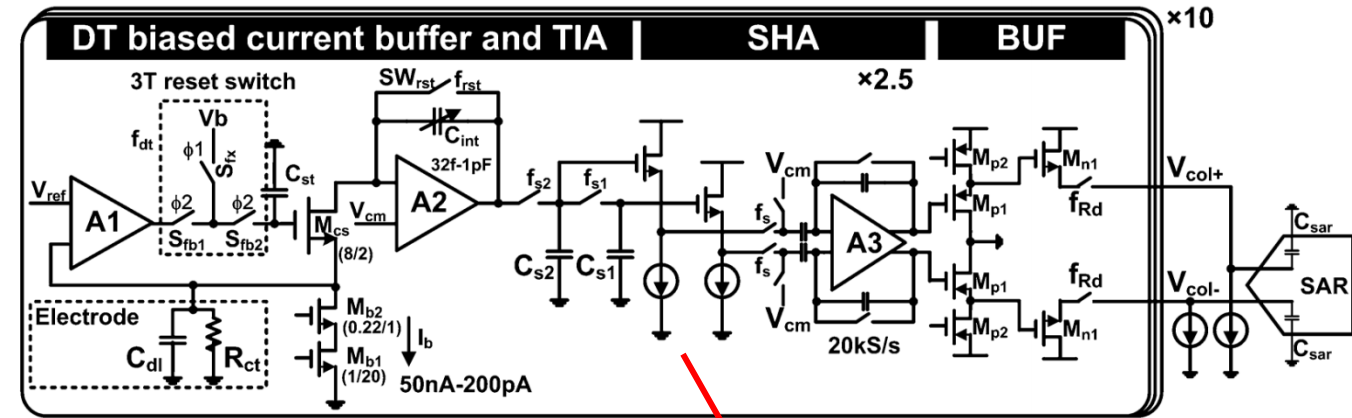
Multimodal Readout Circuits

Voltage and current readout



- 200 pixels
- 16 pixel/mm²

Current-readout channel

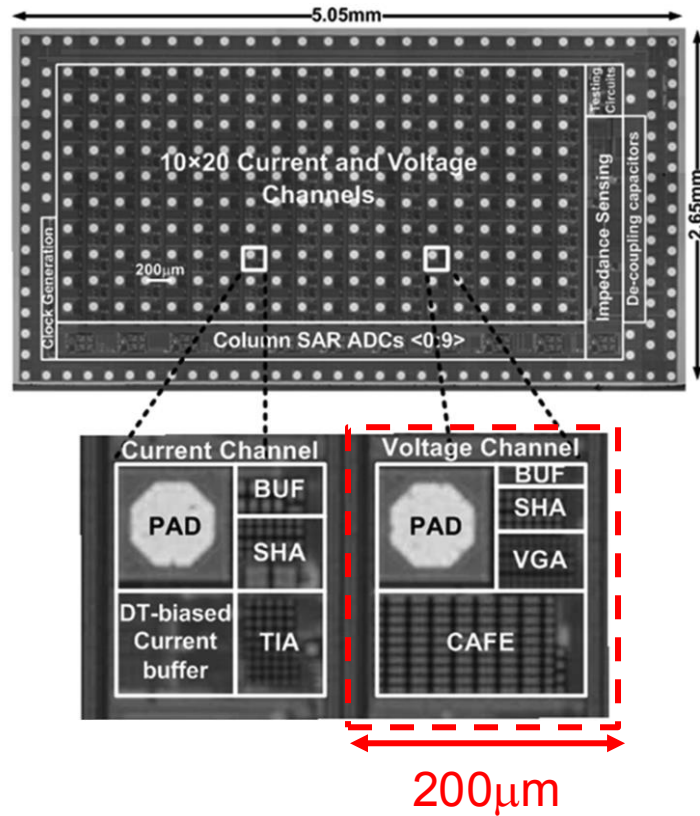


Current buffer & TIA



Multimodal Readout Circuits

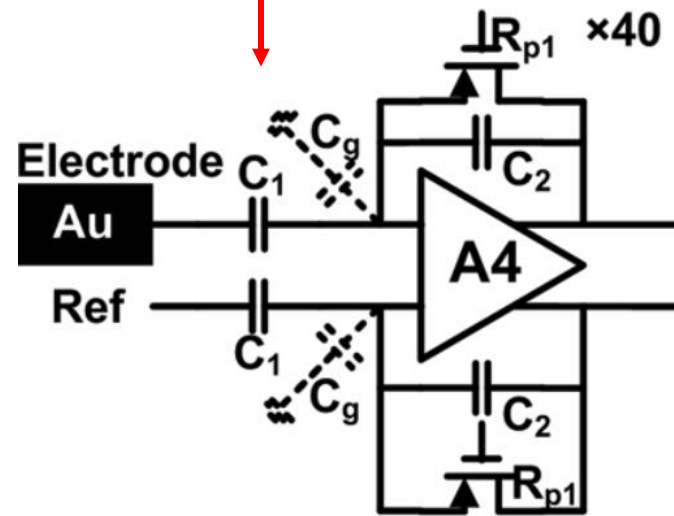
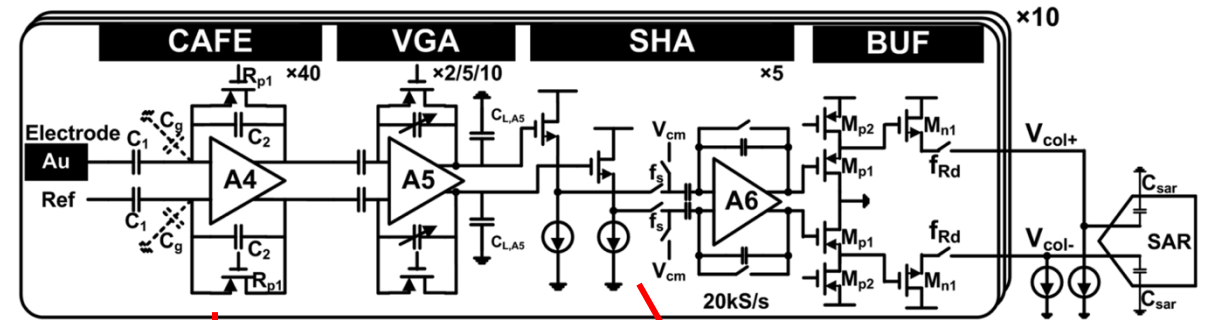
Voltage and current readout



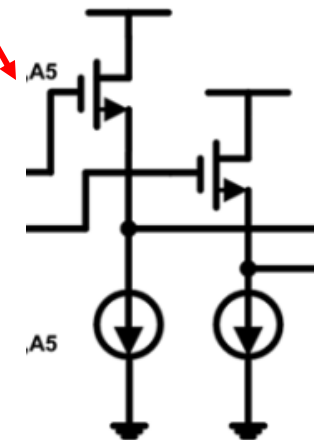
- 200 pixels
- 16 pixel/mm²

Guo TBCAS '16

Voltage-readout channel



Cap-coupled INA



Guo TBCAS '16

